Round-Optimal Threshold Blind Signatures without Random Oracles

Georg Fuchsbauer[1], Fabian Regen[1], and Hoeteck Wee[2]

[1] TU Wien, Austria
[2] NTT Research

Abstract. This paper presents the first round-optimal threshold blind signature without random oracles. Our construction achieves security in the algebraic group model (AGM) for asymmetric pairing groups, and tolerates adaptive corruption of up to $t-1$ signers, where t is the threshold. We improve upon the recent threshold blind signatures of Lehmann, Nazarian and Özbay (EUROCRYPT 2025) and Jarecki and Nazarian (ASIACRYPT 2025) in two ways: we eliminate both the reliance on random oracles and the need for q-type assumptions in the AGM. As a core building block, we introduce a new pairing-based round-optimal blind signature without random oracles, based on the 2-DL assumption in the AGM. Both blind signature schemes achieve communication and computation costs only twice that of the celebrated blind BLS signature.

1 Introduction

Blind signatures [Cha82] define a protocol between a user and an issuer that lets the user obtain a signature on a message so that the issuer does not learn anything about the message-signature pair. Introduced for e-cash [Cha82, CFN90, OO92, Bra94, BCKL09, BFQ21], they have found numerous other applications, such as e-voting [FOO93] and anonymous credentials [Bra94, CL01, CL04, CG08, BL13, FHS15] and also deployed in several industry applications [PCM21, App, Goo, HIP$^+$, Tru]. The security requirements are two-fold: (i) (one-more) unforgeability, wherein a malicious user that runs several concurrent signing sessions with an issuer cannot produce more valid signatures than the number of completed signing sessions, and (ii) blindness, wherein a malicious issuer cannot link a valid message-signature pair to the signing session that produced it.

A highly desirable feature for blind signatures is round-optimality, where the user and issuer are required to only send one message each to complete the signing protocol. Apart from minimizing latency, round-optimal schemes have two important benefits: the signer does not need to maintain any state across signing sessions, and the scheme achieves concurrent security for free. In addition, we focus on schemes that do not require a trusted set-up, which is problematic for real-world deployment.

Technical challenge. The main technical challenge in building round-optimal blind signatures without trusted set-up lies in proving one-more unforgeability, wherein the reduction needs to produce q valid signatures on blinded messages, and then decide which of the $q+1$ valid message-signature pairs provided by the adversary corresponds to an actual forgery, without breaking blindness. Indeed, Fischlin and Schröder [FS10] showed that a broad class of round-optimal blind signature schemes cannot be instantiated in the standard model based on any standard, non-interactive assumption. Existing efficient round-optimal blind signatures address the technical challenge and circumvent the impossibility result in one of several ways:

- Using "one-more" hardness assumptions, which assert that an adversary cannot produce $q+1$ solutions of a problem, given q queries to an oracle for solving the problem. These are *interactive* assumptions, whose structure closely mirrors the one-more unforgeability security requirement.
- Using random oracles [BR93], typically to instantiate non-interactive zero-knowledge proofs of knowledge to extract the message from the blinded signing queries [Fis06, CL03].
- Considering idealized adversaries, notably using the algebraic group model (AGM) [FKL18] together with non-interactive q-type assumptions whose size grows with the number of signing queries. An example is q-DL, which asserts that it's hard to compute y given $g, g^y, g^{y^2}, \ldots, g^{y^q}$.

State of the art. The celebrated pairing-based blind BLS signature [Bol03, BLS01], along with its precursor blind RSA [Cha82], rely on both random oracles and one-more assumptions. Blind BLS is notable for its small signature size and communication: 48 B and 96 B respectively at 128-bit security. Numerous works sought to improve the assumptions used in blind BLS at the cost of larger signatures and communication: these include

(1) schemes without random oracles that use either one-more assumptions, or the AGM plus q-DL, namely [FHS15, FHKS16, HK17, BFR24], where the state of the art in [HK17] achieves signature size 144 B and communication 240 B, and
(2) schemes based on standard assumptions with random oracles [dPK22, AJOR18, HLW23, KRS23], where the state of the art in [KRS23] achieves signature size 447 B and communication 303 B (alternatively, 96 B and 2.2 KB respectively).

Meanwhile, several works showed that the one-more assumptions used in blind signatures can be reduced to q-DL in the AGM [BFP21, JX25]. Henceforth, we will regard these one-more assumptions as a special case of AGM plus q-DL.

Threshold blind signatures. More recently, there has been growing interest in threshold signatures, wherein the signing power is distributed among n parties, ensuring that only a quorum of t issuers can generate a valid signature [Des88, DF90]. This setup is particularly useful in applications where trust is decentralized, such as in blockchain systems or digital voting mechanisms as well as in anonymous credentials to mitigate the devastating costs of corrupted issuing authorities.

Interest in threshold blind signatures has grown recently, and fewer constructions are known. The existing round-optimal schemes [VZK03, LNÖ25, JN25] start by threshold'izing blind BLS via Shamir secret-sharing in the exponent, and as such, inherit the use of random oracles and one-more assumptions of blind BLS. We do not know any round-optimal threshold blind signature without random oracles. In particular, the existing round-optimal blind signature schemes are not threshold'izable, because the signing key is multiplied by fresh randomness during signing.

This work. The goal of this work is to simultaneously eliminate both the reliance on random oracles and the need for q-type assumptions in the AGM from the state-of-the-art blind signatures. We pose the question:

Can we build efficient round-optimal blind signatures —as well as threshold blind signatures— from pairings without random oracles, based on static assumptions in the AGM?

By a static assumption, we mean an assumption whose size is fixed and does not grow with the number of signing queries q, e.g. 1-DL. Recall that static assumptions are provably weaker than q-type assumptions in the AGM [BFL20, JX25] (e.g., we can separate q-DL and $(q+1)$-DL via the meta-reduction technique). Furthermore, using static assumptions could yield better concrete security since we avoid Cheon's attack [Che06]. As such, achieving security under static assumptions in the AGM is a meaningful and worthwhile goal.

Answering the above question requires overcoming two hurdles, along with new techniques for the design and analysis of blind signatures. The first relates to threshold'ization: we need a new design for round-optimal blind signatures without random oracles that does not require multiplying the signing key by fresh randomness during signing. The second relates to avoiding q-type assumptions. In prior blind signatures, as well as analysis of one-more assumptions in the AGM, q-type assumptions are needed to handle recursive queries, where a group element provided by the signing/solution oracle is used as an input to the next query.

Our results. We address the above open problem affirmatively. We present the first pairing-based round-optimal blind signature without random oracles, based on the 2-DL assumption[3] in the AGM with a tight security reduction. Building on this, we obtain the first round-optimal threshold blind signature without random oracles. Our threshold blind signature also relies on the 2-DL assumption in the AGM, and achieves threshold one-more unforgeability (OMUF-1 in [LNÖ25], the strongest possible without session IDs), even against adaptive corruptions with erasures, again with a tight security reduction.

Furthermore, both schemes are concretely efficient, featuring communication and computation costs only twice that of the blind BLS signature, with signature size 96 B and communication 192 B; this improves upon the state of the art described earlier in (1) and (2). We refer to Tables 1 and 2 for a detailed comparison with prior works.

As a secondary contribution, we show that our blind signature schemes support efficient zero-knowledge proofs of knowledge (zkPoK) of message-signature pairs (albeit in the random oracle model); such zkPoK are important for applications such as anonymous credentials [BCC04, CL04, BBS04]. Note that blind BLS does not support efficient zkPoK.

Our approach. We mention a few high-level ideas behind our constructions and how we overcome the aforementioned hurdles, on which we elaborate further in the technical overview. Our starting point is the same as that for the Katsumata-Reichle-Sakai (KRS) blind signature [KRS23], wherein we naively augment the Boneh-Boyen IBE-based (BB) signatures [BB04] with blind issuances. Unfortunately, a malicious user can recover the secret key via a carefully crafted signing query; this attack arises because BB signatures are linear in the signing key, which is also why they are readily threshold'izable. KRS circumvents this attack by having the user provide a non-interactive zero-knowledge proof of knowledge that

[3] More precisely, we rely on the $(2,1)$-DL assumption in AGM, which says that it's hard to compute y given $g_1, g_2, g_1^y, g_1^{y^2}, g_2^y$, where g_1 and g_2 are the respective generators of the two source groups of the pairing.

Table 1: Comparison with blind BLS [Bol03, BLS01] and round-optimal pairing-based blind signatures without random oracles in [FHS15, FHKS16, HK17], where the second line for each scheme shows the concrete sizes for BLS12-381. The column "comm" refers to the total communication for signing protocol; "sig size" refers to the signature size; t_{sign} refers to signing time for the issuer; and t_{verify} refers to the signature verification time. We only include the dominant terms for the timings, notably exponentiations in $\mathbb{G}_1, \mathbb{G}_2$ (denoted by E_1, E_2), and pairings (denoted by P). The column "assumption" takes into account the security proof [BFR24] for the scheme [FHS19] used in [FHS15, FHKS16]. The last column indicates whether blindness is perfect (✓) or computational (×).

Scheme	comm	sig size	t_{sign}	t_{verify}	assumption	perf. blind
blind BLS	$2\mathbb{G}_1$ 96 B	$1\mathbb{G}_1$ 48 B	$1E_1$	$2P$	one-more + ROM	✓
FHS15	$4\mathbb{G}_1 + 1\mathbb{G}_2$ 288 B	$4\mathbb{G}_1 + 1\mathbb{G}_2$ 288 B	$3E_1 + 1E_2$	$7P$	q-type + AGM	×
FHKS16	$6\mathbb{G}_1 + 1\mathbb{G}_2$ 384 B	$7\mathbb{G}_1 + 3\mathbb{G}_2$ 624 B	$5E_1 + 1E_2$	$15P$	q-type + AGM	×
HK17	$5\mathbb{G}_1$ 240 B	$3\mathbb{G}_1$ 144 B	$5E_1$	$5P$	one-more	✓
this work	$4\mathbb{G}_1$ 192 B	$2\mathbb{G}_1$ 96 B	$2E_1$	$2P$	$(2,1)$-DL + AGM	✓

additionally satisfies "online extractability"[4] – this is where they rely on random oracles and incur larger communication. We circumvent the same attack using a novel implicit proof of knowledge, which relies on the AGM for extraction and requires only a single group element. Moreover, an adversary cannot generate accepting proofs for group elements derived from previous signatures. This renders recursive queries useless, and allows us to avoid the use of q-type assumptions.

2 Technical overview

We use bilinear groups of prime order p with an asymmetric pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$. We denote $\mathbb{G}_1$ and $\mathbb{G}_2$ additively and for a fixed generator g_i of $\mathbb{G}_i$, we use "implicit notation", writing $[a]_i$ for the group element $a \cdot g_i$, where $a \in \mathbb{Z}_p$.

Warm-up. As mentioned earlier, we build on the KRS blind signature and BB signatures. We begin with a recap of BB signatures, which satisfies a weak form of selective unforgeability from a static assumption, as well as EUF-CMA in the Generic Group Model (GGM) [ABGW17]. For simplicity, we describe the scheme for signing messages in $\mathbb{Z}_p$ instead of message vectors in $\mathbb{Z}_p^\ell$.

[4] Informally, this means that the knowledge extractor can recover the witness without rewinding the prover.

Table 2: Comparison with threshold blind BLS [LNÖ25, VZK03], Snowblind [CKM+23] and twice-blind threshold blind BLS [JN25]. The column "sig" refers to the signature size; one "round" of communication refers to two moves; "no RO" refers to whether the construction is random oracle free. T-BOMDH and OMDL are "one-more", and thus interactive, assumptions.

Scheme	sig	rounds	no RO	assumption	corruptions
threshold blind BLS	$\mathbb{G}_1$	1	×	T-BOMDH	static
Snowblind	$\mathbb{G}+2\mathbb{Z}_p$	3	×	AGM + DL	static
JN25	$\mathbb{G}_1$	1	×	AGM + OMDL + DDH + SDL	adaptive
this work	$2\mathbb{G}_1$	1	✓	AGM + (2,1)-DL	adaptive

- The public key pk comprises $[x]_T, [w]_1, [u]_1, [w]_2, [u]_2$ for $x, w, u \leftarrow_R \mathbb{Z}_p$ and the secret key sk is $[x]_1$.
- A signature on $m \in \mathbb{Z}_p$ is the pair $[x + s(w + mu)]_1, [s]_1$, for $s \leftarrow_R \mathbb{Z}_p$.
- Signatures can be randomized by replacing s with $s + s'$ for a fresh $s' \leftarrow_R \mathbb{Z}_p$ as follows: add $s' \cdot [(w + mu)]_1$ to the first component and $[s']_1$ to the second component.

To turn the above scheme into a blind signature scheme, consider the following protocol for blind issuance, which is the starting point of the KRS blind signature:

- The user sends a Pedersen commitment $[\hat{m}]_1 := [r]_1 + m \cdot [u]_1$ for $r \leftarrow_R \mathbb{Z}_p$.
- The issuer responds with the "pre-signature" $[x]_1 + s \cdot \big([w]_1 + [\hat{m}]_1\big)$, $[s]_1$ for $s \leftarrow_R \mathbb{Z}_p$.
- The user recovers $[x + s(w + mu)]_1$ using r, and then randomizes the signature.

Even though this scheme satisfies perfect blindness, it is not unforgeable: e.g., an adversary can send $[\hat{m}]_1 := -[w]_1$ and recover the secret key $[x]_1$ from the issuer's answer.

To avoid the above attack, we would like the user to additionally "prove" that $[\hat{m}]_1$ is of the form $[r]_1 + m \cdot [u]_1$. KRS uses an online-extractable non-interactive zero-knowledge proof instantiated using random oracles, following [Fis05, Fis06], and most of the technical novelty in KRS lies in constructing and optimizing such a proof. In particular, naively compiling a Σ-protocol via Fiat-Shamir requires rewinding and does not achieve online-extractability.

Our blind signature scheme. We require the user to provide an implicit "proof" of knowledge of m and r:

- Key generation additionally samples $v \leftarrow_R \mathbb{Z}_p$, and appends $([v]_1, [vu]_1)$ to pk and v to sk.
- The user sends $[\hat{m}]_1$ as before, along with $[\hat{m}']_1 := r \cdot [v]_1 + m \cdot [vu]_1$.
- The issuer rejects if $[\hat{m}']_1 \neq v \cdot [\hat{m}]_1$, and otherwise proceeds as before.
- The user proceeds as before to recover and randomize the signature.

Roughly speaking, we show that in the AGM, we can "extract" m, r from $[\hat{m}]_1$. In the AGM the adversary must compute group elements "algebraically"; concretely, when given the generator $[1]_1$ and $[u]_1$, then together with $[\hat{m}]_1$, it must return a "representation" $r, m \in \mathbb{Z}_p$ with $[\hat{m}] = r \cdot [1]_1 + m \cdot [u]_1$. In our scheme the adversary receives many more group elements, so its representation for $[\hat{m}]_1$ can also depend on these. We show that if extraction of m, r fails, we can solve 2-DL. If it succeeds, one-more unforgeability essentially follows from EUF-CMA of the Boneh-Boyen signature. Perfect blindness follows from the following observations:

- The user's query perfectly hides the message m.
- Given the public key and the user's query, we can check if a pre-signature is well-formed.
- If a pre-signature is well-formed, then the derived signature is a fresh signature on m whose distribution only depends on pk and m.

Threshold blind signature scheme. Threshold'ization is straightforward since the signature is linearly homomorphic with respect to the secret key x. We distribute the secret key amongst a set of n issuers so that any subset of size t can sign:

- Key generation computes t-out-of-n shares $x_1, \dots, x_n$ of x and assigns $[x_i]_1$ to the i'th issuer.[5]
- The elements $[v]_1$, $[v]_2$ and $[vu]_1$ move from pk to the public parameters pp and the issuer uses a pairing to check whether $v \cdot [\hat{m}]_1 = [\hat{m}']_1$. This allows the scheme to use the same v across all issuers, which simplifies the protocol, and comes at the cost of replacing an exponentiation by a pairing during signing.
- The i-th issuer computes a partial pre-signature $[x_i]_1 + s_i \cdot \big([w]_1 + [\hat{m}]_1\big)$, $[s_i]_1$ for $s_i \leftarrow_{\mathrm{R}} \mathbb{Z}_p$.
- Aggregation recovers the partial signatures $[x_i + s_i(w + mu)]_1, [s_i]_1$ using r, combines t partial signatures via Lagrangian interpolation "in the exponent" to a single signature that is valid under the public key $[x]_T$, and randomizes the signature.

An adversary can corrupt any subset C of up to $t-1$ issuers, upon which it learns the corresponding $[x_i]_1$. With static corruptions, the adversary must pick C at the beginning of the security game. With adaptive corruptions, as achieved in our work,[6] the adversary can adaptively decide which issuer to corrupt.

Overview of AGM security proofs. We prove one-more unforgeability of both our blind signature scheme as well as the threshold variant under the 2-DL assumption in the AGM. Recall that in AGM, any group element produced by an adversary is accompanied by a "representation" that expresses the group element as a linear combination of all the group elements it has seen so far. We follow the blueprint laid out in [FKL18, BFL20], which comprises two steps:

- Step 1: We first analyze "symbolic" security wherein we treat uniformly random group elements chosen by honest parties and the challenger as indeterminates (similar to a proof in the GGM). This allows us to associate every group element in the security game with a multivariate polynomial in the indeterminates. Polynomials corresponding to group elements produced by the adversary are derived from their accompanying AGM representations. We prove structural properties about these polynomials.

[5] Our proof of adaptive security crucially uses the fact that the issuers' signing key shares are group elements and not scalars; this makes our proof simpler than analogous results for threshold (blind) BLS [BL22, DR24, JN25], where the signing key share is a scalar.

[6] We assume erasures, which for our protocol means that the adversary does not get to see the random scalars s used in the previously queried pre-signatures.

– Step 2: Next, we move from the symbolic to the computational setting by carefully embedding the secret exponent y from the 2-DL challenge into the indeterminates via a random linear function of y; this turns every multivariate polynomial in the indeterminates of Step 1 into a univariate polynomial. The AGM reduction can keep track of these polynomials, using the representations the algebraic adversary provides. We then use symbolic security of the scheme (shown in Step 1) to argue that an algebraic adversary that breaks one-more-unforgeability must produce a non-zero univariate polynomial h (of low degree) such that $h(y) = 0$, upon which we can solve for y. To argue that h is non-zero, we crucially exploit the fact that the coefficient of y in the linear function is perfectly hidden from the adversary.

We make two additional high-level observations about the above blueprint:

– An important quantity is the maximal total degree D of the multivariate polynomials in Step 1 that correspond to group elements provided to the adversary. This is because the reduction needs to compute univariate polynomials of degree D in y "in the exponent" in Step 2. In our proof, the bound for D is 2 for $\mathbb{G}_1$ and 1 for $\mathbb{G}_2$, which means the 2-DL challenge $[y]_1, [y^2]_1, [y]_2$ suffices to evaluate these polynomials "in the exponent". In contrast, the AGM security proofs for prior round-optimal blind signatures without random oracles require D that grows with the number of signing queries, hence the need for q-type assumptions.
– In Step 2, the reduction knows the random linear function in y corresponding to every honestly generated group element, and can therefore compute all of these group elements. In the case of our threshold blind signature wherein secret keys are group elements, this means that the reduction can compute the secret keys for every issuer. This allows us to handle adaptive corruptions.

One-more unforgeability. We provide a brief overview of Step 1 for our blind signature scheme. We will associate the uniformly random exponents x, w, u, v with indeterminates X, W, U, V. Suppose an adversary makes q queries to the signing oracle.

(i) First, consider polynomials $\hat{m}, \hat{m}' \in \mathbb{Z}_p[X, U, V, W]$ that correspond to a signing query $[\hat{m}]_1, [\hat{m}']_1$. (E.g., given the $\mathbb{G}_1$ elements of the public key $([w]_1, [u]_1, [v]_1, [vu]_1)$, the adversary accompanies its first query $[\hat{m}]_1$ by an AGM representation $(\alpha, \beta, \gamma, \delta, \zeta)$, i.e., $[\hat{m}]_1 = \alpha[1]_1 + \beta[w]_1 + \gamma \cdot [u]_1 + \delta[v]_1 + \zeta[vu]_1$; the polynomial associated with $[\hat{m}]_1$ is thus $\alpha + \beta W + \gamma U + \delta V + \zeta VU$.)
If $V\hat{m} \equiv \hat{m}'$ (as an equality of polynomials) then $\hat{m} = mU + r$ for some $m, r \in \mathbb{Z}_p$. This follows from the fact that $\hat{m}'$ must lie in the span of V and UV, since these are the only multiples of V known to the adversary. We note that the fact that $\hat{m}$ has total degree 1 allows us to bound D by 2. By a straight-forward induction, we can associate all q queries to the signing oracle with scalars $m_1, r_1, \ldots, m_q, r_q \in \mathbb{Z}_p$. At this point, security reduces to EUF-CMA of the BB signature.

(ii) Next, we leverage the verification equation of the BB signature scheme to argue that any forgery on a message m^* must satisfy $m^* \in \{m_1, \ldots, m_q\}$. This implies that the adversary can produce valid signatures on at most q distinct messages.

We refer to Lemma 3 and 4 respectively for more details for each of these two statements. Step 2 follows the blueprint described earlier: depending on whether the adversary violates (i) or (ii), the reduction has a different way to solve for y.

Table 3: Comparison of zkPoK for BBS [TZ23] where $k \le \ell$ is the number of *non*-disclosed components. The terms in the timings include exponentiations in $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$ (denoted by E_1, E_2 and E_T), membership tests in $\mathbb{G}_1$ (denoted by M_1) and pairings (denoted by P).

BS scheme	proof size	prover time	verifier time
BBS	$2\lvert\mathbb{G}_1\rvert + (k+3)\lvert\mathbb{Z}_p\rvert$	$(\ell+5)E_1$	$2P + 2M_1 + (\ell+3)E_1$
this work	$2\lvert\mathbb{G}_1\rvert + (k+2)\lvert\mathbb{Z}_p\rvert$	$P + (\ell+3)E_1 + (k+1)E_2$	$2P + 2M_1 + E_T + (\ell+2)E_2$

The proof of one-more-unforgeability for the t-out-of-n threshold scheme has the same structure, but differs in part (ii) of Step 1: here, we show that for any forgery m^* there must be at least $t - |C|$ partial signing queries that correspond to m^*, where C is the set of corrupted issuers. This is formalized in Lemma 6, and the full proof appears in Section 5.2.

Signing vectors. Our scheme supports signing message vectors $\mathbf{m} = \big(\mathbf{m}[1], \ldots, \mathbf{m}[\ell]\big) \in \mathbb{Z}_p^\ell$. The scalar u in the public key elements $[u]_1$ and $[u]_2$ expands to a vector $\mathbf{u} = \big(\mathbf{u}[1], \ldots, \mathbf{u}[\ell]\big) \in \mathbb{Z}_p^\ell$, and mu in the signature gets replaced by $\sum_i \mathbf{m}[i]\mathbf{u}[i]$. Since the issuer does not need to know $\mathbf{u}$, we can put $[\mathbf{u}]_1, [\mathbf{u}]_2$ in the public parameters which are re-used across many issuers. The user can check that these parameters are well-formed using a pairing, and as long as this verification check passes, blindness of our scheme holds even if the parameters have been generated by the adversary.

Efficient zkPok. To build anonymous credentials from our scheme, we give a proof of selective disclosure of message components. Concretely, we present efficient zero-knowledge proofs of knowledge of message-signature pairs where messages are vectors. Suppose all but k components of the message are known to the verifier. Our proof size is $2|\mathbb{G}_1| + (k+2)|\mathbb{Z}_p|$, which is shorter than the zkPoK based on BBS Signatures [TZ23], whose proofs are of size $2|\mathbb{G}_1| + (k+3)|\mathbb{Z}_p|$. For a full comparison we refer to Table 3.

Organization. We describe our blind signature scheme in Section 4 and the threshold blind signature scheme in Section 5. We describe the efficient zkPoK in Appendix B.

Additional related works. We have constructions of round-optimal blind signatures from standard assumptions without random oracles via complexity leveraging [GG14, GRS+11, KNYY21], which are mainly of theoretical interest and not practical. For instance, the optimized protocol in [GG14] achieves signature size 6.5 KB and communication 100 KB. In addition, these constructions require sub-exponential hardness or post-quantum hardness to carry out complexity leveraging. The Boneh-Boyen-Shacham (BBS) signature [BBS04, CL04] can be converted to a round-optimal blind signature, wherein the final signature comprises an online-extractable zero-knowledge proof of knowledge (zkPoK) of a plain BBS signature [TZ23]. The latter would require random oracles, and incur substantial communication overhead like in the KRS blind signature. The work of [SAB+19] describes a round-optimal pairing-based threshold blind signature with random oracles, starting from the Pointcheval-Sanders signature [PS16, CL04]; formally proving security would again require an online-extractable zkPoK, like in the blind KRS signature.

3 Preliminaries

Let $\mathbb{N} := \{1,2,\ldots\}$ denote the natural numbers.For $n \in \mathbb{N}$ define $[n] := \{1,\ldots,n\}$. Let $\mathbb{Z}$ denote the integers. Let $\mathbb{Z}_p := \mathbb{Z}/p\mathbb{Z}$ and let $\mathbb{Z}_p[X]$ denote the polynomial ring in indeterminate X over $\mathbb{Z}_p$. We denote polynomial equality by $\equiv$. By monomial we refer to *primitive monomial*, that is, a power product of indeterminates.

We denote vectors by $\mathbf{m} = (\mathbf{m}[1],\ldots,\mathbf{m}[|\mathbf{m}|])$ and for $J \subseteq [|\mathbf{m}|]$, we let $\mathbf{m}_J := (\mathbf{m}[i])_{i\in J}$. Let A be a set and $\mathbf{m} \in A^a$ and $\mathbf{n} \in A^b$ be vectors then $\mathbf{m}\|\mathbf{n} \in A^{a+b}$ denotes the vector where the first a components are $\mathbf{m}$ and the next b components are $\mathbf{n}$.

When denoting a set containing elements and vectors, we abuse notation and define $\{x,\mathbf{y}\} := \{x,\mathbf{y}[1],\ldots,\mathbf{y}[|\mathbf{y}|]\}$. For $\mathbf{x},\mathbf{y} \in \mathbb{Z}_p^\ell$ define $\langle \mathbf{x},\mathbf{y}\rangle := \sum \mathbf{x}[i]\mathbf{y}[i]$. For a vector $\mathbf{v} \in (\mathbb{Z}_p[X])^\ell$ of polynomials we define $\langle \mathbf{y},\mathbf{v}\rangle$ as the polynomial $x \mapsto \sum \mathbf{y}[i]\mathbf{v}[i](x)$.

Let S be a finite set, by $s \leftarrow_{\mathrm{R}} S$ we denote that s is chosen uniformly from S. For a (possibly randomized) algorithm A we write $y \leftarrow A(x)$ to denote that y is the output of A on input x.

Lemma 1 (Schwartz-Zippel). *Let p be prime and $P \in \mathbb{Z}_p[X_1,\ldots,X_n]$ be a non-zero polynomial of total degree d. Then*

$$\Pr_{r_1,\ldots,r_n \leftarrow_{\mathrm{R}} \mathbb{Z}_p}\left[P(r_1,\ldots,r_n) = 0\right] \le \frac{d}{p}.$$

The next lemma [BFL20, Lemma 2.1] has become a standard tool in AGM proofs.

Lemma 2. *Let P be a non-zero multivariate polynomial in $\mathbb{Z}_p[X_1,\ldots,X_n]$ of total degree d. If we define $Q(Y) \in \big(\mathbb{Z}_p[W_1,\ldots,W_m,V_1,\ldots,V_n]\big)[Y]$ as*

$$Q(Y) := P\big(W_1Y + V_1,\ldots,W_nY + V_n\big),$$

then the coefficient of maximal degree of Q is a polynomial in $\mathbb{Z}_p[W_1,\ldots,W_n]$ of degree d.

Bilinear Groups. Our scheme is defined over a bilinear group $grp = (\mathbb{G}_1,\mathbb{G}_2,\mathbb{G}_T,p,g_1,g_2,e)$ where $\mathbb{G}_i$ is a group of prime order p for $i \in \{1,2,T\}$, g_1 and g_2 are generators of $\mathbb{G}_1$ and $\mathbb{G}_2$ respectively, and $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ is a bilinear map such that $g_T := e(g_1,g_2)$ generates $\mathbb{G}_T$. We denote $\mathbb{G}_1$ and $\mathbb{G}_2$ additively and use "implicit" representation of group elements: for $x \in \mathbb{Z}_p$ and $i \in \{1,2\}$ let $[x]_i := x \cdot g_i$. This notation will naturally be extended to vectors: for $\mathbf{x} \in \mathbb{Z}_p^\ell$ let $[\mathbf{x}]_i := \big([\mathbf{x}[1]]_i,\ldots,[\mathbf{x}[\ell]]_i\big) \in \mathbb{G}_i^\ell$. For a vector $\mathbf{x} \in \mathbb{Z}_p^\ell$ and $y \in \mathbb{Z}_p$ we define the pairings $e([\mathbf{x}]_1,[y]_2) := \big(e([\mathbf{x}[1]]_1,[y]_2),\ldots,e([\mathbf{x}[\ell]]_1,[y]_2)\big) \in \mathbb{G}_T^\ell$ and $e([y]_1,[\mathbf{x}]_2) := \big(e([y]_1,[\mathbf{x}[1]]_2),\ldots,e([y]_1,[\mathbf{x}[\ell]]_2)\big) \in \mathbb{G}_T^\ell$ componentwise.

Shamir secret sharing. We use Shamir's secret sharing for $x \in \mathbb{Z}_p$ and $t < n \in \mathbb{N}$ as follows: $\mathsf{Share}(x,n,t)$ samples elements $a_1,\ldots,a_{t-1} \in \mathbb{Z}_p$, defines $f(Z) := x + \sum_{i=1}^{t-1} a_i Z^i$ and outputs the shares $\big(f(1),\ldots,f(n)\big)$.

The Algebraic Group Model (AGM). Introduced by Fuchsbauer, Kiltz and Loss [FKL18], the algebraic group model is a model that sits between the generic group model (GGM) and the standard model. Whereas a generic algorithm merely sees encodings of group elements and accesses group operations via oracles, an algebraic algorithm gets full access to group elements and operations but when it outputs a group element it also has to provide a representation with regard to all previously seen group elements.

Definition 1. *An algorithm $\mathcal{A}$ is* algebraic *if whenever it outputs a group element $h \in \mathbb{G}$ it also provides scalars λ_i such that $h = \sum_i \lambda_i g_i$ where $(g_i)_{i \in I}$ are the elements that $\mathcal{A}$ has previously seen, for appropriate (finite) I.*

Assumption in the AGM. The (q_1, q_2)-DL assumption [Lip12] in a bilinear group states that for $y \leftarrow_{\mathrm{R}} \mathbb{Z}_p$ no efficient adversary that is given $[y^i]_1$ for $i \leq q_1$ and $[y^i]_2$ for $i \leq q_2$ can find y.

We consider the special case for $q_1 = 2$ and $q_2 = 1$.

Definition 2. *Let grp be a bilinear group, then the $(2,1)$-DL assumption is (ε, τ)-hard in grp if for all algorithms $\mathcal{A}$ running in at most time τ it holds that $\mathsf{Adv}^{(2,1)\text{-DL}}_{\mathcal{A}} \leq \varepsilon$, where the advantage of $\mathcal{A}$ is defined as follows:*

$$\mathsf{Adv}^{(2,1)\text{-DL}}_{\mathcal{A}} := \Pr\left[\begin{array}{l} y \leftarrow_{\mathrm{R}} \mathbb{Z}_p, \\ y' \leftarrow \mathcal{A}(grp, [y]_1, [y^2]_1, [y]_2) \end{array} : y' = y\right].$$

3.1 Blind Signatures

Our definitions will be with respect to a fixed bilinear group *grp*, which will be in line both with proofs in the AGM and implementations in practice.

Definition 3. *A (round-optimal)* blind signature scheme Σ *consists of the following PPT algorithms:*

- $\mathsf{Setup}()$*: outputs public parameters pp, which implicitly define the message space $\mathcal{M}$.*
- $\mathsf{Keygen}(pp)$*: outputs a key pair (sk, pk).*
- $\mathsf{VerifyPK}(pk)$*: a deterministic algorithm that ouputs 1 if pk is a valid public key and 0 otherwise.*
- $(\mathsf{User}, \mathsf{Signer}, \mathsf{Derive})$ *is a triple of algorithms that describe the components of the interactive protocol run by the user and the issuer. First the user runs $\mathsf{User}(pk, m)$ taking a message and outputs a first protocol message $\hat{M}$ and a state st. Then the issuer runs $\mathsf{Signer}(sk, \hat{M})$ and outputs a* pre-signature *$\hat{\sigma}$. Lastly the user runs $\mathsf{Derive}(st, \hat{\sigma})$ outputting a signature σ on m.*
- $\mathsf{Verify}(pk, m, \sigma)$*: a deterministic algorithm that outputs 1 if σ is a valid signature for m under pk and 0 otherwise.*

Remark 1. We describe our scheme with public parameters *pp* to achieve better concrete efficiency. Here, *pp* can be reused across multiple issuers. This does not constitute a trusted set-up however, since blindness holds as long as *pp* passes the check specified by $\mathsf{VerifyPK}$. In particular, we could remove Setup by incorporating Setup and *pp* into KeyGen and *pk* respectively. Note that $\mathsf{VerifyPK}$ was defined as a separate algorithm since verification of an issuer's public key only has to be done once by the user, and it simplifies the definition of the blindness game with respect to adversarially generated keys.

We follow the standard definitions of *correctness, unforgeability* and *(malicious-issuer) blindness.*

Definition 4. *A blind signature scheme* Σ *(as in Def. 3) is* correct *if for all* $pp \leftarrow \mathsf{Setup}()$*, all keys* $(sk, pk) \leftarrow \mathsf{Keygen}(pp)$*, all messages* $m \in \mathcal{M}$ *and*

$$\sigma \leftarrow \mathsf{Derive}(st, \mathsf{Signer}(sk, \hat{m})) \;\; for \;\; (\hat{m}, st) \leftarrow \mathsf{User}(pk, m),$$

it holds that $\mathsf{Verify}(pk, m, \sigma) = 1$.

Definition 5. *A blind signature scheme* Σ *is* (ε, τ)*-unforgeable if for all algorithms* $\mathcal{A}$ *running in time at most* τ *that have access to a signing oracle it holds that* $\mathsf{Adv}^{\mathsf{OMUF}}_{\Sigma,\mathcal{A}} \leq \varepsilon$*, where the advantage is defined via the game* $\mathsf{OMUF}_{\Sigma,\mathcal{A}}$ *defined in Figure 1:*

$$\mathsf{Adv}^{\mathsf{OMUF}}_{\Sigma,\mathcal{A}} := \Pr\left[\mathsf{OMUF}_{\Sigma,\mathcal{A}} = 1\right].$$

$\underline{\mathsf{OMUF}_{\Sigma,\mathcal{A}}}$	$\underline{\mathcal{O}^{\mathsf{Signer}}(\hat{m})}$
$pp \leftarrow \mathsf{Setup}()$	$q := q+1$
$(sk, pk) \leftarrow \mathsf{Keygen}(pp)$	output $\mathsf{Signer}(sk, \hat{m})$
$q := 0$	
$(m_i, \sigma_i)_{i=1}^{q+1} \leftarrow \mathcal{A}^{\mathcal{O}^{\mathsf{Signer}}(\cdot)}(pk)$	
if $\exists i, j \in [q+1] : i \neq j \wedge m_i = m_j$: output 0	
if $\exists i \in [q+1] : \mathsf{Verify}(pk, m_i, \sigma_i) = 0$: output 0	
output 1	

Fig. 1: The unforgeability game $\mathsf{OMUF}_{\Sigma,\mathcal{A}}$ for a blind signature scheme Σ.

Definition 6. *A blind signature scheme* Σ *is* $(\varepsilon, \tau)-$blind *if for all algorithms* $\mathcal{A}$ *running in time at most* τ *it holds that* $\mathsf{Adv}^{\mathsf{BLND}}_{\Sigma,\mathcal{A}} \leq \varepsilon$*, where the advantage is defined via the game* $\mathsf{BLND}_{\Sigma,\mathcal{A}}$ *defined in Figure 2:*

$$\mathsf{Adv}^{\mathsf{BLND}}_{\Sigma,\mathcal{A}} := \left|\Pr\left[\mathsf{BLND}_{\Sigma,\mathcal{A}} = 1\right] - \tfrac{1}{2}\right|.$$

Σ *is* perfectly blind *if the above holds for* $\varepsilon = 0$ *and for unbounded adversaries* $\mathcal{A}$.

3.2 Threshold Blind Signatures

Threshold blind signatures combine both properties of blind signatures, where the issuer cannot link valid signature-message pairs to signing sessions, and threshold signatures, where a set of t out of n issuers has to sign a message in order to create a valid signature. Our definitions will be with respect to a fixed bilinear group *grp*, which will be in line both with proofs in the AGM and implementations in practice. The following definitions were adapted from Lehmann et al. [LNÖ25] to the setting of 1-round issuingand incorporate adaptive corruptions following Jarecki and Nazarian [JN25].

$\underline{\mathsf{BLND}_{\Sigma,\mathcal{A}}}$
$b \leftarrow_R \{0,1\}$
$(pk, m_0, m_1, st) \leftarrow \mathcal{A}()$
if $\mathsf{VerifyPK}(pk) = 0$: output 0
for $i \in \{0,1\}$:
    $(\hat{m}_i, st_i) \leftarrow \mathsf{User}(pk, m_i)$
$(\hat{\sigma}_b, \hat{\sigma}_{1-b}, st) \leftarrow \mathcal{A}(st, \hat{m}_b, \hat{m}_{1-b})$
for $i \in \{0,1\}$:
    $\sigma_i \leftarrow \mathsf{Derive}(st_i, \hat{\sigma}_i)$
if $\exists i \in \{0,1\} : \mathsf{Verify}(pk, m_i, \sigma_i) = 0$
    $\sigma_0 := \perp, \sigma_1 := \perp$
$b' \leftarrow \mathcal{A}(st, \sigma_0, \sigma_1)$
if $b = b'$: output 1
output 0

Fig. 2: The blindness game $\mathsf{BLND}_{\Sigma,\mathcal{A}}$ for a blind signature scheme Σ [HKL19].

Definition 7. *A (1-round) threshold blind signature scheme* Σ *consists of the following PPT algorithms:*

- $\mathsf{Setup}()$*: outputs public parameters pp.*
- $\mathsf{Keygen}(n, t)$*: outputs a public key pk and n key shares* $sk_1, \ldots, sk_n$*.*
- $\mathsf{VerifyPK}(pk)$*: a deterministic algorithm that ouputs* 1 *if pk is a valid public key and* 0 *otherwise.*
- $(\mathsf{User}, \mathsf{Signer}, \mathsf{Aggregate})$ *is a triple of algorithms that describe the components of the interactive protocol run by the user and each issuer. First the user runs* $\mathsf{User}(pk, m)$ *taking a message and outputs a first protocol message* $\hat{M}$ *and a state st. Then the issuer i runs* $\mathsf{Signer}(sk_i, \hat{M})$ *and outputs a pre-signature* $\hat{\sigma}_i$*. Lastly, the user runs* $\mathsf{Aggregate}(st, I, (\hat{\sigma}_i)_{i\in I})$ *where* $I \subseteq [n]$ *with* $|I| \geq t$*, combining pre-signatures to a signature* σ *on* m*.*
- $\mathsf{Verify}(pk, m, \sigma)$*: a deterministic algorithm that outputs* 1 *if* σ *is a valid signature for* m *under pk and* 0 *otherwise.*

Next, we define *correctness*, *unforgeability* and *blindness*.

Definition 8. *A threshold blind signature scheme* Σ *is* correct *if for all* $pp \leftarrow \mathsf{Setup}()$*, all keys* $\big((sk_1, \ldots, sk_n), pk\big) \leftarrow \mathsf{Keygen}(pp)$*, all messages* $m \in \mathcal{M}$*, all* $I \subseteq [n]$ *with* $|I| \geq t$ *and* $\sigma \leftarrow \mathsf{Aggregate}\big(st, I, \big(\mathsf{Signer}(sk_i, \hat{m})\big)_{i\in I}\big)$ *for* $(\hat{m}, st) \leftarrow \mathsf{User}(pk, m)$ *it holds that* $\mathsf{Verify}(pk, m, \sigma) = 1$*.*

To define unforgeability, we first need to define a function $\mathsf{count}()$ which determines the maximum number of signatures an adversary can "trivially" compute. Observe that an adversary corrupting a set of C issuers with $|C| < t$ can obtain a valid signature on a message $\mathbf{m}$ by asking $t - |C|$ of the non-corrupted issuers for a partial signature on $\mathbf{m}$. We will consider a function $\rho : [q] \to [n]$ that maps the i-th query to the $\rho(i)$-th issuer. Then an adversary can obtain a valid signature on $\mathbf{m}$ when for the set $Q \subseteq [q]$ of queries for $\mathbf{m}$ we have $|\rho(Q)| \geq t - |C|$. The function count will determine the largest number k of disjoint sets $Q_1,\dots,Q_k$ such that for each set Q_i we have $|\rho(Q_i)| \geq t - |C|$. That is, count counts the maximum number of signatures the adversary can obtain by following the protocol.

Definition 9. *Let $\rho : [q] \to [n]$, $t \in [n]$, $C \subseteq [n]$ and define*

$$\mathsf{count}(\rho,t,C) := \max\left\{k \;\middle|\; \exists J_1,\dots,J_k \subseteq [q] : \left(J_i \cap J_j = \emptyset \textit{ for } i \neq j\right) \wedge \left(\forall i \in [k] : |\rho(J_i) \cup C| \geq t\right)\right\}.$$

Definition 10. *A threshold blind signature scheme Σ is (ε,τ)-unforgeable if for all algorithms $\mathcal{A}$ running in time at most τ and having access to a signing oracle and a corruption oracle it holds that $\mathsf{Adv}^{\mathsf{tOMUF}}_{\Sigma,\mathcal{A}} \leq \varepsilon$, where the advantage is defined via the game $\mathsf{tOMUF}_{\Sigma,\mathcal{A}}$ given in Figure 3:*

$$\mathsf{Adv}^{\mathsf{tOMUF}}_{\Sigma,\mathcal{A}} := \Pr\left[\mathsf{tOMUF}_{\Sigma,\mathcal{A}} = 1\right].$$

$\underline{\mathsf{tOMUF}_{\Sigma,\mathcal{A}}}$	$\underline{\mathcal{O}^{\mathsf{Signer}}(j,\hat{m})}$
$pp \leftarrow \mathsf{Setup}()$	$q := q+1$
$(n,t) \leftarrow \mathcal{A}(pp)$	$\rho(q) := j$
if $n \leq t$: output 0	output $\mathsf{Signer}(sk_j,\hat{m})$
$(sk_1,\dots,sk_n,pk) \leftarrow \mathsf{Keygen}(n,t)$	
$C := \emptyset$, $q := 0$, $\rho := \mathbf{0}$	$\underline{\mathcal{O}^{\mathsf{Corrupt}}(j)}$
$(m_i,\sigma_i)_{i=1}^k \leftarrow \mathcal{A}^{\mathcal{O}^{\mathsf{Signer}}(\cdot,\cdot),\mathcal{O}^{\mathsf{Corrupt}}(\cdot)}(pk)$	$C := C \cup \{j\}$
if $\|C\| \geq t$: output 0	output sk_j
if $\exists i,j \in [k] : i \neq j \wedge m_i = m_j$: output 0	
if $\exists i \in [k] : \mathsf{Verify}(pk,m_i,\sigma_i) = 0$: output 0	
if $k \leq \mathsf{count}(\rho,t,C)$: output 0	
output 1	

Fig. 3: The unforgeability game $\mathsf{tOMUF}_{\Sigma,\mathcal{A}}$ for a threshold blind signature scheme Σ. Informally, the adversary succeeds if it manages to output valid signatures for at least $\mathsf{count}(\rho,t,C)+1$ distinct messages, where count (Definition 9) defines how many signed messages an honest adversary could have obtained.

Remark 2 (Comparison with [LNÖ25]). Our definition is based on OMUF-1, the strongest notion for protocols without session IDs. Lehmann et al. use a slightly different definition of count, equivalence of both notions is shown in Appendix A. Note that we consider a *randomizable* scheme, which cannot achieve the *strong* unforgeability. The winning condition $(m_i,\sigma_i) \neq (m_j,\sigma_j)$ is thus changed to $m_i \neq m_j$. We consider adaptive corruptions with erasures, wherein the adversary can make adaptive queries to a corruption oracle that returns the issuer's secret key.

Definition 11. *A threshold blind signature scheme is (ε,τ)-*blind *if for all algorithms $\mathcal{A}$ running in time at most τ it holds that $\mathsf{Adv}^{\mathsf{tBLND}}_{\Sigma,\mathcal{A}} \leq \varepsilon$, where the advantage is defined via the game $\mathsf{tBLND}_{\Sigma,\mathcal{A}}$ defined in Figure 4:*

$$\mathsf{Adv}^{\mathsf{tBLND}}_{\Sigma,\mathcal{A}} := \left|\Pr\left[\mathsf{tBLND}_{\Sigma,\mathcal{A}} = 1\right] - \tfrac{1}{2}\right|.$$

A scheme is perfectly blind *if it is $(0,\infty)$-blind.*

$\underline{\mathsf{tBLND}_{\Sigma,\mathcal{A}}}$
$b \leftarrow_R \{0,1\}$
$(pk, m_0, m_1, S_0, S_1, st) \leftarrow \mathcal{A}()$
if $\mathsf{VerifyPK}(pk) = 0$: output 0
$(\hat{m}_b, st_b) \leftarrow \mathsf{User}(pk, m_b)$
$(\hat{m}_{1-b}, st_{1-b}) \leftarrow \mathsf{User}(pk, m_{1-b})$
$\big((\hat{\sigma}_{b,j})_{j\in S_0}, (\hat{\sigma}_{1-b,j})_{j\in S_1}, st\big) \leftarrow \mathcal{A}(st, \hat{m}_b, \hat{m}_{1-b})$
$\sigma_b \leftarrow \mathsf{Aggregate}(st_b, S_0, (\hat{\sigma}_{b,j})_{j\in S_0})$
$\sigma_{1-b} \leftarrow \mathsf{Aggregate}(st_{1-b}, S_1, (\hat{\sigma}_{1-b,j})_{j\in S_1})$
if $\exists i \in \{0,1\}$ $\mathsf{Verify}(pk, m_i, \sigma_i) = 0$:
  $\sigma_0 := \perp; \sigma_1 := \perp$
$b' \leftarrow \mathcal{A}(st, \sigma_0, \sigma_1)$
if $b = b'$: output 1
output 0

Fig. 4: The blindness game $\mathsf{tBLND}_{\Sigma,\mathcal{A}}$ for a threshold blind signature scheme Σ.

4 Our Blind Signature Scheme

We refer the reader to Section 2 for a technical overview. Our scheme is defined in Fig. 5.

4.1 Correctness and Blindness

First we show that our scheme is correct.

Theorem 1 (Correctness). *The scheme in Fig. 5 is correct.*

Proof. By Derive and Signer we get

$$\begin{aligned}[z]_1 &= [\hat{z}]_1 - r[\hat{s}]_1 + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1\rangle\big)\\ &= [x]_1 + \hat{s}[w]_1 + \hat{s}[r]_1 + \hat{s}\langle \mathbf{m}, [\mathbf{u}]_1\rangle - r[\hat{s}]_1 + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1\rangle\big)\\ &= [x]_1 + (\hat{s}+\tilde{s})\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1\rangle\big).\end{aligned}$$

Since $s = \hat{s} + \tilde{s}$ we get $e\big([z]_1, [1]_2\big) = e\big([s]_1, [w]_2 + \langle \mathbf{m}, [\mathbf{u}]_2\rangle\big) \cdot [x]_T$. □

We will establish that our scheme is perfectly blind by specifying simulator S which outputs a distribution D that is independent of the bit b chosen in the experiment, such that D is distributed identically to the view of $\mathcal{A}$.

Theorem 2 (Blindness). *The scheme in Fig. 5 is perfectly blind.*

$\underline{\mathsf{Setup}()}$
$\mathbf{u} \leftarrow_R \mathbb{Z}_p^\ell, w \leftarrow_R \mathbb{Z}_p$
output $pp := ([\mathbf{u}]_1, [\mathbf{u}]_2, [w]_1, [w]_2)$

$\underline{\mathsf{Keygen}(pp)}$
$x, v \leftarrow_R \mathbb{Z}_p$
set $sk := ([x]_1, v, pp)$
$pk := ([x]_T, pp, [v]_1, [v]_2, [v\mathbf{u}]_1)$
output (sk, pk)

$\underline{\mathsf{User}(pk, \mathbf{m})}$
$r \leftarrow_R \mathbb{Z}_p$
$[\hat{m}]_1 := [r]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle$
$[\hat{m}']_1 := r[v]_1 + \langle \mathbf{m}, [v\mathbf{u}]_1 \rangle$
output $([\hat{m}]_1, [\hat{m}']_1), st := (r, \mathbf{m})$

$\underline{\mathsf{Signer}(sk, [\hat{m}]_1, [\hat{m}']_1)}$
if $[\hat{m}']_1 \neq v[\hat{m}]_1$: output $\perp$
$s \leftarrow_R \mathbb{Z}_p$
output $\big([x]_1 + s([w]_1 + [\hat{m}]_1), [s]_1\big)$

$\underline{\mathsf{Derive}(st = (r, \mathbf{m}), [\hat{z}]_1, [\hat{s}]_1)}$
$\tilde{s} \leftarrow_R \mathbb{Z}_p$
$[z]_1 := [\hat{z}]_1 - r[\hat{s}]_1 + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big)$
$[s]_1 := [\hat{s}]_1 + [\tilde{s}]_1$
$\sigma := ([z]_1, [s]_1)$
if $\mathsf{Verify}(pk, \mathbf{m}, \sigma) = 1$: output σ
output $\perp$

$\underline{\mathsf{Verify}\big(pk, \mathbf{m}, ([z]_1, [s]_1)\big)}$
output $e\big([z]_1, [1]_2\big) \stackrel{?}{=} e\big([s]_1, [w]_2 + \langle \mathbf{m}, [\mathbf{u}]_2 \rangle\big) \cdot [x]_T$

$\underline{\mathsf{VerifyPK}(pk)}$
if $e([v]_1, [\mathbf{u}]_2) \neq e([v\mathbf{u}]_1, [1]_2)$ or
$e([1]_1, [\mathbf{u}]_2) \neq e([\mathbf{u}]_1, [1]_2)$ or
$e([1]_1, [w]_2) \neq e([w]_1, [1]_2)$ or
$e([1]_1, [v]_2) \neq e([v]_1, [1]_2)$: output 0
output 1

Fig. 5: Our blind signature scheme defined over *grp* with message space $\mathcal{M} = \mathbb{Z}_p^\ell$. A signature on $\mathbf{m}$ is of the form $\big([x + s(w + \langle \mathbf{m}, \mathbf{u} \rangle)]_1, [s]_1\big)$ for $s \leftarrow_R \mathbb{Z}_p$. If we wish to avoid any setup or structured *pp*, then we can have KeyGen sample *pp* (cf. Remark 1).

Proof. Consider the blindness experiment. We specify an (unbounded) simulator S that is independent of b, which runs $\mathcal{A}$ and outputs a distribution D that is distributed identically to $\mathcal{A}$'s view in the experiment. The adversary outputs elements $pp = ([w]_1, [\mathbf{u}]_1, [w']_1, [\mathbf{u}']_1)$ and $pk = ([x]_T, [v]_1, [v']_2, [\mathbf{y}]_1)$ and two messages $\mathbf{m}_0, \mathbf{m}_1$. Simulator S generates the distribution D of $([\hat{m}_0]_1, [\hat{m}_1]_1, [\hat{m}_0']_1, [\hat{m}_1']_1, \sigma_0, \sigma_1)$ as follows: It first runs $\mathsf{VerifyPK}(pk)$ and aborts if the public key was not valid. This simulates the experiment. Otherwise, the simulator S samples $[\hat{m}_0]_1, [\hat{m}_1]_1 \leftarrow \mathbb{G}_1$ uniformly and lets $[\hat{m}_i']_1 := v[\hat{m}_i]_1$ for $i \in \{0,1\}$. These elements are sent to $\mathcal{A}$, which replies with $\hat{\sigma}_i \in \mathbb{G}_1^2$ for $i \in \{0,1\}$.

For $i \in \{0,1\}$ let $r_i \in \mathbb{Z}_p$ be the unique value such that $\hat{m}_i = r_i + \langle \mathbf{m}_0, \mathbf{u} \rangle$. If $\exists i \in \{0,1\} : \mathsf{Derive}((r_i, \mathbf{m}_0), \hat{\sigma}_i) = \perp$ then S sets $\sigma_1 = \sigma_2 = \perp$. Now $[\hat{m}_i]_1$ are distributed identically to $\mathcal{A}$'s view for $i \in \{0,1\}$ due to the way r is chosen in User, and the $[\hat{m}_i']_1$ are uniquely determined. Therefore D is distributed identically to $\mathcal{A}$'s view.

On the other hand, if $\forall i \in \{0,1\} : \mathsf{Derive}(\hat{\sigma}_i) \neq \perp$ then S samples $s_1, s_2 \leftarrow \mathbb{Z}_p$ and sets $\sigma_i = ([x + s_i(w + \langle \mathbf{m}_i, \mathbf{u} \rangle)]_1, [s_i]_1$ for $i \in \{0,1\}$. Since $\mathsf{VerifyPK}(pk) = 1$, we have that $w = w'$, $v = v'$, $\mathbf{u} = \mathbf{u}'$ and $\mathbf{y} = v\mathbf{u}$. Therefore, the σ_i are valid signatures on $\mathbf{m}_i$ that are distributed identically to outputs of Derive. As mentioned above, $[\hat{m}_i]_1$ and $[\hat{m}'_i]_1$ for $i \in \{0,1\}$ are distributed identically to $\mathcal{A}$'s view. Therefore D is distributed identically to $\mathcal{A}$'s view.

In either case D is distributed identically to $\mathcal{A}$'s view. Since $\mathcal{S}$ is independent of b this implies that the scheme is perfectly blind. □

4.2 Unforgeability

As mentioned in the overview in Section 2, the proof has two main steps.

Step 1: Symbolic Security. To show unforgeability in the AGM we first analyze some polynomial equalities. This essentially shows that an adversary cannot symbolically forge a signature. For this we will associate uniformly random exponents $x, w, \mathbf{u}, v, s_1, \ldots, s_q$ with indeterminates (denoted by the corresponding capitalized letters $X, W, \mathbf{U}, V, S_1, \ldots, S_q$). For example, when an adversary, after having previously seen the elements $[1]_1$, $[w]_1$, $[\mathbf{u}]_1$, $[v]_1$, $[v\mathbf{u}]_1$, outputs a group element $[\hat{m}_1]_1$ together with a *representation* $(\alpha, \beta, \boldsymbol{\gamma}, \delta, \boldsymbol{\zeta})$ such that

$$[\hat{m}_1]_1 = \alpha[1]_1 + \beta[w]_1 + \langle \boldsymbol{\gamma}, [\mathbf{u}]_1 \rangle + \delta[v]_1 + \langle \boldsymbol{\zeta}, [v\mathbf{u}]_1 \rangle,$$

the reduction defines the polynomial $\hat{m}_1 \in \mathbb{Z}_p[X, W, \mathbf{U}, V, S_1, \ldots, S_q]$ as

$$\hat{m}_1 := \alpha + \beta W + \langle \boldsymbol{\gamma}, \mathbf{U} \rangle + \delta V + \langle \boldsymbol{\zeta}, V\mathbf{U} \rangle.$$

Therefore $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$ can be interpreted as "in the representation of $[\hat{m}_i]_1$ provided by $\mathcal{A}$ all but the coefficients of $[1]_1$ and $[\mathbf{u}]_1$ were zero".

Later, the reduction is going to embed a DL solution y into these uniformly random exponents using the DL challenge $[y]_1$. Therefore, if the reduction knew y (and it knows how y was embedded into the exponents), evaluating $\hat{m}_1$ at these exponents yields $\log[\hat{m}_1]_1 \in \mathbb{Z}_p$. This motivates the slight abuse of notation (since one would expect $\hat{m}_1$ to denote the logarithm of $[\hat{m}_1]_1$).

Let $\hat{m}_i \in \mathbb{Z}_p[X, W, \mathbf{U}, V, S_1, \ldots, S_q]$ and for $i \leq q$ inductively define $z_i \in \mathbb{Z}_p[X, W, \mathbf{U}, V, S_1, \ldots, S_q]$ as

$$z_i := X + S_i(W + \hat{m}_i). \tag{1}$$

The following lemma shows that the "proof" of representation guarantees that $[\hat{m}_i]_1$ was formed correctly.

Lemma 3 (Structure of $\hat{m}_i$). *Let $i \leq q$ and let $\hat{m}_i, \hat{m}'_i \in \mathsf{span}\{1, W, V, \mathbf{U}, V\mathbf{U}, S_1, \ldots, S_{i-1}, z_1, \ldots, z_{i-1}\}$. If $v \cdot \hat{m}_i \equiv \hat{m}'_i$, then $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$.*

Proof. We proceed by induction on $i \in [q]$.

<u>*Base case.*</u> Here, $V \cdot \hat{m}_1 \equiv \hat{m}'_1 \in \mathsf{span}\{1, W, V, \mathbf{U}, V\mathbf{U}\}$. The only multiples of V appearing on the RHS are $V, V\mathbf{U}$. Therefore, $V \cdot \hat{m}_1 \in \mathsf{span}\{V, V\mathbf{U}\}$ and thus $\hat{m}_1 \in \mathsf{span}\{1, \mathbf{U}\}$.

Induction step. By the induction hypothesis, $\hat{m}_1, \ldots, \hat{m}_{i-1} \in \mathsf{span}\{1, \mathbf{U}\}$, which means

$$z_1, \ldots, z_{i-1} \in \mathsf{span}\big\{X, S_1, \ldots, S_{i-1}, S_1 W, \ldots, S_{i-1} W, S_1 \mathbf{U}, \ldots, S_{i-1}\mathbf{U}\big\}.$$

In particular, none of the monomials of $z_1, \ldots, z_{i-1}$ are multiples of V. We also have

$$V \cdot \hat{m}_i \equiv \hat{m}'_i \in \mathsf{span}\big\{1, W, V, V\mathbf{U}, \mathbf{U}, S_1, \ldots, S_{i-1}, z_1, \ldots, z_{i-1}\big\}.$$

Again, the only multiples of V appearing on the RHS are $V, V\mathbf{U}$. Therefore, $V \cdot \hat{m}_i \in \mathsf{span}\{V, V\mathbf{U}\}$ and thus $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$. □

The next lemma states that whenever an adversary provides a forgery for which the verification equation holds as a polynomial equality, then the forgery has to be on a message that was queried previously.

Lemma 4 (Structure of forgeries). *Let z_i be defined as in (1) where $\hat{m}_i = r_i + \langle \mathbf{m}_i, \mathbf{U}\rangle$ for $r_i \in \mathbb{Z}_p$ and $\mathbf{m}_i \in \mathbb{Z}_p^\ell$. Suppose the polynomials $z^*, s^* \in \mathsf{span}\{1, W, V, \mathbf{U}, V\mathbf{U}, S_1, \ldots, S_q, z_1, \ldots, z_q\}$ and the vector $\mathbf{m}^* \in \mathbb{Z}_p^\ell$ satisfy the polynomial equality*

$$X + s^*\big(W + \langle \mathbf{m}^*, \mathbf{U}\rangle\big) \equiv z^*. \tag{2}$$

Then it holds that $\mathbf{m}^* \in \{\mathbf{m}_1, \ldots, \mathbf{m}_q\}$.

Proof. Let $\boldsymbol{\theta}, \boldsymbol{\eta} \in \mathbb{Z}_p^{2\ell+3}$ and $\alpha_i, \beta_i, \gamma_i, \delta_i \in \mathbb{Z}_p$ for $i \in [q]$ be such that

$$\begin{aligned} z^* &= \Big\langle \boldsymbol{\theta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U} \Big\rangle + \sum_i \alpha_i S_i + \beta_i z_i, \\ s^* &= \Big\langle \boldsymbol{\eta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U} \Big\rangle + \sum_i \gamma_i S_i + \delta_i z_i. \end{aligned} \tag{3}$$

Recall $z_i = X + S_i(r_i + W + \langle \mathbf{m}_i, \mathbf{U}\rangle)$. Expanding (2) by (3) yields

$$X + \big(\langle \boldsymbol{\eta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U}\rangle + \sum_i \gamma_i S_i + \delta_i z_i\big)\big(W + \langle \mathbf{m}^*, \mathbf{U}\rangle\big) = \langle \boldsymbol{\theta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U}\rangle + \sum_i \alpha_i S_i + \beta_i z_i. \tag{4}$$

Comparing coefficients of $S_i W^2$ we get :

$$\delta_1 = \cdots = \delta_q = 0.$$

With this, keeping only scalar multiples of X, $S_i W$, $S_i \mathbf{U}$ (i.e. comparing coefficients of these monomials) in (4) yields:

$$X + \big(\sum_i \gamma_i S_i\big)\big(W + \langle \mathbf{m}^*, \mathbf{U}\rangle\big) = \sum_i \beta_i \big(X + S_i W + \langle \mathbf{m}_i, S_i \mathbf{U}\rangle\big).$$

Comparing coefficients of $S_i W$ yields $\gamma_i = \beta_i$. With this, comparing coefficients of $S_i \mathbf{U}$ yields $\beta_i \mathbf{m}^* = \beta_i \mathbf{m}_i$, which means that

$$\beta_i = 0 \vee \mathbf{m}^* = \mathbf{m}_i \quad \forall i \in [q].$$

Comparing coefficients of X yields $\sum_i \beta_i = 1$. Therefore, there exists $i \in [q]$ such that $\beta_i \neq 0$, implying $\mathbf{m}^* \in \{\mathbf{m}_1, \ldots, \mathbf{m}_q\}$. □

Step 2: Moving to the AGM. Provided these technical lemmata, we are now able to tackle unforgeability of our blind signature scheme in the AGM.

Theorem 3 (Unforgeability in the AGM). *Let grp be a bilinear group of order p. Let $\mathcal{A}$ be an algebraic adversary against* OMUF *security of the blind signature scheme Σ for grp defined in Fig. 5 that makes q signature queries and runs in time τ. Then there exists an algorithm $\mathcal{R}$ solving $(2,1)$-DL with respect to grp, running in time at most $\tau + \mathcal{O}(q \log p)$ such that*

$$\mathsf{Adv}^{\mathsf{OMUF}}_{\Sigma,\mathcal{A}} \leq \mathsf{Adv}^{(2,1)\text{-DL}}_{\mathcal{R}} + \frac{3}{p}.$$

Proof. Assume that $\mathcal{A}$ makes q *valid* signature queries $[\hat{m}]_1, [\hat{m}']_1$, that is, queries for which $[\hat{m}']_1 = v \cdot [\hat{m}]_1$ holds. This assumption is without loss of generality since for any adversary $\mathcal{A}$ there exists an adversary $\mathcal{A}'$ that behaves like $\mathcal{A}$ but checks whether a query is valid by checking $e\big([\hat{m}']_1, [1]_2\big) \stackrel{?}{=} e\big([\hat{m}]_1, [v]_2\big)$ and discards the query if it is invalid.

We will proceed as follows: first we describe how the reduction $\mathcal{R}$ chooses parameters and answers Signer queries by embedding the $(2,1)$-DL solution y into group elements. We will then analyze two cases that depend on $\mathcal{A}$'s queries, where in each case $\mathcal{R}$ will use $\mathcal{A}$ to derive a polynomial h that is non-zero with high probability, such that $h(y) = 0$. The reduction can then solve for y with high probability by computing the roots of h. At last we will fully describe $\mathcal{R}$.

Consider a $(2,1)$-DL challenge $[1]_1, [1]_2, [y]_1, [y^2]_1, [y]_2$ given to the reduction. To embed solution y into the public parameters, the public key, and the randomness used during Signer queries, the reduction samples values $\tilde{y}, \bar{y} \leftarrow \mathbb{Z}_p$ and then embeds y in $[\bar{y} + \tilde{y}y]$. Here $\tilde{y}$ ensures that y is perfectly hidden from $\mathcal{A}$, whereas $\bar{y}$ ensures that $\tilde{y}$ stays perfectly hidden from $\mathcal{A}$, which will be of relevance later. The reduction $\mathcal{R}$ proceeds as follows:

- For the public parameters and public key $\mathcal{R}$ samples $\tilde{x}, \bar{x}, \tilde{v}, \bar{v}, \tilde{w}, \bar{w} \leftarrow \mathbb{Z}_p$, $\tilde{\mathbf{u}}, \bar{\mathbf{u}} \leftarrow \mathbb{Z}_p^\ell$ and implicitly programs:

  $$x = \bar{x} + \tilde{x} \cdot y, \quad v = \bar{v} + \tilde{v} \cdot y, \quad w = \bar{w} + \tilde{w} \cdot y, \quad \mathbf{u} = \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y.$$

  Using the $(2,1)$-DL challenge it can then output the corresponding group elements, e.g. $[v]_1 = [\bar{v}]_1 + \tilde{v}[y]_1$ for the adversary. In the case of the public key it uses the pairing e to transform the secret key $[x]_1$ into $[x]_T$.
- On a Signer query with polynomial representation (derived from the adversary's AGM representation) $\hat{m}_i = r_i + \langle \mathbf{m}_i, \mathbf{U} \rangle \in \mathsf{span}\{1, \mathbf{U}\}$ the reduction samples $\tilde{s}_i, \bar{s}_i$ and programs

  $$s_i = \bar{s}_i + \tilde{s}_i \cdot y.$$

  It then computes

  $$\begin{aligned} z_i &= x + s_i(w + \hat{m}_i) \\ &= x + s_i(w + r_i + \langle \mathbf{m}_i, \mathbf{u} \rangle) \\ &= \bar{x} + \tilde{x} \cdot y + (\bar{s}_i + \tilde{s}_i \cdot y)(\bar{w} + \tilde{w} \cdot y + r_i + \langle \mathbf{m}_i, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y \rangle) \\ &= \bar{x} + \bar{s}_i(\bar{w} + r_i + \langle \mathbf{m}_i, \bar{\mathbf{u}} \rangle) + \big(\tilde{x} + \bar{s}_i(\tilde{w} + \langle \mathbf{m}_i, \tilde{\mathbf{u}} \rangle) + \tilde{s}_i(\bar{w} + r_i + \langle \mathbf{m}_i, \tilde{\mathbf{u}}' \rangle)\big) \cdot y + \tilde{s}_i(\tilde{w} + \langle \mathbf{m}_i, \tilde{\mathbf{u}} \rangle) \cdot y^2 \end{aligned}$$

  and outputs the answer $[z_i]_1, [s_i]_1$ using the $(2,1)$-DL challenge $[y]_1$ and $[y^2]_1$.

Note that since $\tilde{x}$ and $\bar{x}$ are chosen uniformly, the resulting public key is distributed identically as in the security game. The same argument applies to the other parameters and the signatures.

We thus showed how $\mathcal{R}$ chooses the parameters and answers signing queries whenever the representation $\mathcal{A}$ provides satisfies $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$. Next we will consider the case that $\mathcal{A}$ does not provide such a representation on all Signer queries.

<u>*Case 1.*</u> There exists $i \in [q]$ such that $\hat{m}_i \notin \mathsf{span}\{1, \mathbf{U}\}$. Let i be the smallest index for which this is the case. By the above the reduction is able to answer the previous $i-1$ signature queries. Next, using the representations $\mathcal{A}$ provides for $[\hat{m}_i]_1, [\hat{m}'_i]_1$, $\mathcal{R}$ computes a polynomial $P_1 \in \mathbb{Z}_p[X, W, \mathbf{U}, V, S_1, \ldots, S_{i-1}]$ corresponding to $V \cdot \hat{m}_i - \hat{m}'_i$. In particular at query i the previously seen elements comprise $[1]_1, [w]_1, [\mathbf{u}]_1, [v]_1, [v\mathbf{u}]_1$ and the elements $[s_j]_1, [z_j]_1$ for $j < i$. For a representation

$$[\hat{m}_i]_1 = \alpha[1]_1 + \beta[w]_1 + \langle \boldsymbol{\gamma}, [\mathbf{u}]_1 \rangle + \delta[v]_1 + \langle \boldsymbol{\zeta}, [v\mathbf{u}]_1 \rangle + \sum_{j<i} \eta_j [s_j]_1 + \theta_j [z_j]_1,$$

provided by $\mathcal{A}$ the reduction defines the polynomial $\hat{m}_i \in \mathbb{Z}_p[X, W, \mathbf{U}, V, S_1, \ldots, S_{i-1}]$ as

$$\hat{m}_i := \alpha + \beta W + \langle \boldsymbol{\gamma}, \mathbf{U} \rangle + \delta V + \langle \boldsymbol{\zeta}, V\mathbf{U} \rangle + \sum_{j<i} \eta_j S_j + \theta_j \big(X + S_j (W + \hat{m}_j) \big).$$

The polynomial $\hat{m}'_i$ is defined analogously. Since i is the smallest index such that $\hat{m}_i \notin \mathsf{span}\{1, \mathbf{U}\}$ we have that $\deg \hat{m}_j \leq 1$ for $j < i$. Therefore $\deg \hat{m}_i \leq 2$ and $\deg \hat{m}'_i \leq 2$. The polynomial $P_1 := V \cdot \hat{m}_i - \hat{m}'_i$ has the following properties:

(i) P_1 has total degree (at most) 3 (since $\deg \hat{m}_i, \deg \hat{m}'_i \leq 2$ and thus $\deg(V \cdot \hat{m}_i - \hat{m}'_i) \leq 3$).

(ii) P_1 is a non-zero polynomial, due to Lemma 3 and $\hat{m}_i \notin \mathsf{span}\{1, \mathbf{U}\}$.

(iii) $P_1(\bar{x} + \tilde{x} \cdot y, \bar{w} + \tilde{w} \cdot y, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y, \bar{v} + \tilde{v} \cdot y, \bar{s}_1 + \tilde{s}_1 \cdot y, \ldots, \bar{s}_{i-1} + \tilde{s}_{i-1} \cdot y) = 0$, since $v \cdot [\hat{m}_i]_1 = [\hat{m}'_i]_1$ for a valid query.

Next, $\mathcal{R}$ defines the polynomial $h \in \mathbb{Z}_p[Z]$ given by

$$h(Z) = P_1(\bar{x} + \tilde{x} \cdot Z, \bar{w} + \tilde{w} \cdot Z, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot Z, \bar{v} + \tilde{v} \cdot Z, \bar{s}_1 + \tilde{s}_1 \cdot Z, \ldots, \bar{s}_{i-1} + \tilde{s}_{i-1} \cdot Z).$$

By Lemma 2 and (ii), the leading coefficient of h is a non-zero polynomial in $\mathbb{Z}_p[\tilde{X}, \tilde{W}, \tilde{\mathbf{U}}, \tilde{V}, \tilde{S}_1, \ldots, \tilde{S}_{i-1}]$ of degree at most 3. Since $\tilde{x}, \tilde{w}, \tilde{\mathbf{u}}, \tilde{v}, \tilde{s}_1, \ldots, \tilde{s}_{i-1}$ are perfectly hidden from the view of $\mathcal{A}$, applying the Schwartz-Zippel lemma (Lemma 1) yields that h is non-zero with probability at least $1 - 3/p$. From (iii), we know that $h(y) = 0$. Moreover, due to (i), h has degree at most 3. Therefore, $\mathcal{R}$ can solve for y by computing the roots of the polynomial h.

Next we will consider how $\mathcal{R}$ behaves when Case 1 does not occur. We already established that $\mathcal{R}$ can answer all Signer queries correctly. Let's therefore consider that $\mathcal{A}$ wins the OMUF game.

<u>*Case 2.*</u> $\hat{m}_1, \ldots, \hat{m}_q \in \mathsf{span}\{1, \mathbf{U}\}$ and $\mathcal{A}$ wins OMUF. Let $\hat{m}_i = r_i + \langle \mathbf{m}_i, \mathbf{u} \rangle$ and suppose $\mathcal{A}$ provides valid signatures on $q+1$ distinct messages $\mathbf{m}^*_1, \ldots, \mathbf{m}^*_{q+1}$. Then one of these messages $\mathbf{m}^*$ must satisfy $\mathbf{m}^* \notin \{\mathbf{m}_1, \ldots, \mathbf{m}_q\}$. Let $([z^*]_1, [s^*]_1)$ denote the signature for $\mathbf{m}^*$, which satisfies the verification equation. Using the representations $\mathcal{A}$ provides for $[z^*]_1, [s^*]_1$, the reduction computes a polynomial $P_2 \in \mathbb{Z}_p[X, W, \mathbf{U}, V, S_1, \ldots, S_q]$ corresponding to $X + s^*(W + \langle \mathbf{m}^*, \mathbf{U} \rangle) - z^*$ with the following properties:

(i) P_2 has total degree (at most) 3. This is due to the fact that s^* and z^* have total degree (at most) 2.

(ii) P_2 is a non-zero polynomial, due to Lemma 4 and $\mathbf{m}^* \notin \{\mathbf{m}_1, \ldots, \mathbf{m}_q\}$.

(iii) $P_2(\bar{x} + \tilde{x} \cdot y, \bar{w} + \tilde{w} \cdot y, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y, \bar{v} + \tilde{v} \cdot y, \bar{s}_1 + \tilde{s}_1 \cdot y, \ldots, \bar{s}_q + \tilde{s}_q \cdot y) = 0$, since a valid signature satisfies the verification equation.

At this point, we can proceed as in Case 1 to define a polynomial h that due to (ii) is non-zero with probability at least $1 - 3/p$, due to (i) has degree at most 3 and due to (iii) it holds that $h(y) = 0$. The reduction can therefore solve for y by computing the roots of h.

The reduction $\mathcal{R}$. We will now describe $\mathcal{R}$ in more detail. It runs a copy of $\mathcal{A}$ in the unforgeability game OMUF and proceeds as follows:

- First, it computes pp, pk as described above and sends them to $\mathcal{A}$.
- For each signature query $([\hat{m}_i]_1, [\hat{m}'_i]_1)$ it checks whether the representation $\mathcal{A}$ provides for $[\hat{m}_i]_1$ satisfies $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$. If so, it proceeds to compute $([z_i]_1, [s_i]_1)$ as described above. If not, it aborts and proceeds as in Case 1 to solve for y.
- When $\mathcal{A}$ outputs valid signatures on $q+1$ distinct messages, $\mathcal{R}$ proceeds as in Case 2 to solve for y.

We analyze the probability of $\mathcal{R}$ solving $(2,1)$-DL. Let W denote the event that $\mathcal{R}$ wins the $(2,1)$-DL experiment, and C_1, C_2 denote the events that Case 1 or Case 2 occurs respectively. Since $\Pr[C_1] + \Pr[C_2] = 1$ we showed

$$\begin{aligned}
\mathsf{Adv}_{\mathcal{R}}^{(2,1)\text{-DL}} &= \Pr[W \mid C_1]\Pr[C_1] + \Pr[W \mid C_2]\Pr[C_2] \\
&\geq (1 - \tfrac{3}{p})\Pr[C_1] + (1 - \tfrac{3}{p})\mathsf{Adv}_{\Sigma,\mathcal{A}}^{\mathsf{OMUF}}\Pr[C_2] \\
&\geq (1 - \tfrac{3}{p})\mathsf{Adv}_{\Sigma,\mathcal{A}}^{\mathsf{OMUF}} \\
&\geq \mathsf{Adv}_{\Sigma,\mathcal{A}}^{\mathsf{OMUF}} - \tfrac{3}{p}.
\end{aligned}$$

Lastly, $\mathcal{R}$ runs $\mathcal{A}$, which takes time τ, it answers q signing queries and picks public parameters and the public key in time $\mathcal{O}(q \log p)$, and it factors a polynomial of degree at most 3 in time $\mathcal{O}(\log p)$. In total $\mathcal{R}$'s running time is in $\tau + \mathcal{O}(q \log p)$. □

5 Our Threshold Blind Signature Scheme

We refer the reader to Section 2 for a technical overview. Our threshold blind signature scheme is defined in Figure 6. Note that the user only needs to run User once and can broadcast the query to all intended issuers.

5.1 Correctness and Blindness

We will first establish that our threshold blind signature scheme is correct.

Theorem 4 (Correctness). *The scheme in Fig. 6 is correct.*

$\underline{\mathsf{Setup}()}$

$\mathbf{u} \leftarrow_{\mathrm{R}} (\mathbb{Z}_p)^\ell, w \leftarrow_{\mathrm{R}} \mathbb{Z}_p, v \leftarrow_{\mathrm{R}} \mathbb{Z}_p$

$pp := ([\mathbf{u}]_1, [\mathbf{u}]_2, [w]_1, [w]_2, [v]_1, [v]_2, [v\mathbf{u}]_1)$

output pp

$\underline{\mathsf{Keygen}(pp)}$

$x \leftarrow_{\mathrm{R}} \mathbb{Z}_p$

$(x_1, \ldots, x_n) \leftarrow \mathsf{Share}(x, t, n)$

for $i \in [n]$: $sk_i := ([x_i]_1, pp)$

$pk := ([x]_T, pp)$

output $(sk_1, \ldots, sk_n, pk)$

$\underline{\mathsf{User}(pk, \mathbf{m})}$

$r \leftarrow_{\mathrm{R}} \mathbb{Z}_p$

$[\hat{m}]_1 := [r]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle$

$[\hat{m}']_1 := r[v]_1 + \langle \mathbf{m}, [v\mathbf{u}]_1 \rangle$

output $([\hat{m}]_1, [\hat{m}']_1), st := (r, \mathbf{m})$

$\underline{\mathsf{Signer}(i, sk_i, [\hat{m}]_1, [\hat{m}']_1)}$

if $e\big([\hat{m}']_1, [1]_2\big) \neq e\big([\hat{m}]_1, [v]_2\big)$:

  output $\perp$

$s \leftarrow_{\mathrm{R}} \mathbb{Z}_p$

output $\big([x_i]_1 + s([w]_1 + [\hat{m}]_1), [s]_1\big)$

$\underline{\mathsf{Aggregate}(st = (r, \mathbf{m}), I, ([\hat{z}_i]_1, [\hat{s}_i]_1)_{i \in I})}$

for $i \in I$:

  $[z_i]_1 := [\hat{z}_i]_1 - r[\hat{s}_i]_1$

  $[s_i]_1 := [\hat{s}_i]_1$

Let λ_i be the i-th Lagrange coefficient for the set I.

$\tilde{s} \leftarrow_{\mathrm{R}} \mathbb{Z}_p$

$[z]_1 := \sum_{i \in I} \lambda_i [z_i]_1 + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big)$

$[s]_1 := \sum_{i \in I} \lambda_i [s_i]_1 + [\tilde{s}]_1$

if $\mathsf{Verify}(pk, \mathbf{m}, ([z]_1, [s]_1)) = 1$:

  output $\sigma := ([z]_1, [s]_1)$

output $\perp$

$\underline{\mathsf{Verify}(pk, \mathbf{m}, ([z]_1, [s]_1))}$

output $e\big([z]_1, [1]_2\big) \stackrel{?}{=} e\big([s]_1, [w]_2 + \langle \mathbf{m}, [\mathbf{u}]_2 \rangle\big) \cdot [x]_T$

$\underline{\mathsf{VerifyPK}(pk)}$

if $e([v]_1, [\mathbf{u}]_2) \neq e([v\mathbf{u}]_1, [1]_2)$ or

$e([1]_1, [\mathbf{u}]_2) \neq e([\mathbf{u}]_1, [1]_2)$ or

$e([1]_1, [w]_2) \neq e([w]_1, [1]_2)$ or

$e([1]_1, [v]_2) \neq e([v]_1, [1]_2)$: output 0

output 1

Fig. 6: Our (n, t)-threshold blind signature scheme defined over *grp* with message space $\mathcal{M} = \mathbb{Z}_p^\ell$. A signature on $\mathbf{m}$ is of the form $\big([x + s(w + \langle \mathbf{m}, \mathbf{u} \rangle)]_1, [s]_1\big)$ for $s \leftarrow_{\mathrm{R}} \mathbb{Z}_p$.

Proof. By the definition of Aggregate and Signer we get

$$\begin{aligned}
[z]_1 &:= \sum_{i \in I} \lambda_i [z_i]_1 + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big) \\
&= \sum_{i \in I} \lambda_i [\hat{z}_i]_1 - \lambda_i r [\hat{s}_i]_1 + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big) \\
&= \sum_{i \in I} \lambda_i [x_i]_1 + \lambda_i \hat{s}_i\big([w]_1 + [r]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big) - \lambda_i r [\hat{s}_i]_1 + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big) \\
&= \sum_{i \in I} \lambda_i [x_i]_1 + \lambda_i \hat{s}_i\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big) + \tilde{s}\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big),
\end{aligned}$$

since $[x]_1 = \sum_i \lambda_i [x_i]_1$ and $s_i = \hat{s}_i$ we have

$$= [x]_1 + \big(\sum_{i \in I} \lambda_i s_i + \tilde{s}\big)\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big).$$

Now since $s = \sum_{i\in I} \lambda_i s_i + \tilde{s}$ (as defined in $\mathsf{Aggregate}$), we get $e([z]_1, [1]_2) = e([s]_1, [w]_2 + \langle \mathbf{m}, [\mathbf{u}]_2\rangle) \cdot [x]_T$. $\square$

Next we will show that our threshold blind signature scheme is perfectly blind. This will be done similarly to Theorem 2.

Theorem 5 (Blindness). *The scheme in Fig. 6 is perfectly blind.*

Proof. We specify an (unbounded) simulator S that is independent of b, which runs $\mathcal{A}$ and outputs a distribution D that is distributed identically to $\mathcal{A}$'s view in the blindness experiment. The adversary $\mathcal{A}$ outputs elements $pp = ([w]_1, [v]_1, [\mathbf{u}]_1, [\mathbf{y}]_1, [w']_2, [v']_2, [\mathbf{u}']_2)$ and $pk = [x]_T$, two messages $\mathbf{m}_0, \mathbf{m}_1$ and two sets $S_0, S_1 \subseteq [n]$. We define the distribution D of $([\hat{m}_0]_1, [\hat{m}_1]_1, [\hat{m}'_0]_1, [\hat{m}'_1]_1, \sigma_0, \sigma_1)$ generated by the simulator S. It first runs $\mathsf{VerifyPK}(pk)$ and aborts if the public key is invalid. This simulates the experiment.

Otherwise, the simulator S samples $[\hat{m}_0]_1, [\hat{m}_1]_1 \leftarrow_{\mathrm{R}} \mathbb{G}_1$ uniformly and lets $[\hat{m}'_i]_1 := v[\hat{m}_i]_1$ for $i \in \{0,1\}$. These elements are sent to $\mathcal{A}$, which replies with $\hat{\sigma}_{i,j} \in \mathbb{G}_1^2$ for $i \in \{0,1\}$ and $j \in S_i$.

For $i \in \{0,1\}$ let $r_i \in \mathbb{Z}_p$ be the unique value such that $\hat{m}_i = r_i + \langle \mathbf{m}_0, \mathbf{u}\rangle$. Now if

$$\exists i \in \{0,1\} : \mathsf{Aggregate}((r_i, \mathbf{m}_0), S_i, (\hat{\sigma}_{i,j})_{j\in S_i}) = \perp,$$

then S sets $\sigma_1 = \sigma_2 = \perp$. We have that uniformly chosen $[\hat{m}_0]_1, [\hat{m}_1]_1$ are distributed identically to $\mathcal{A}$'s view in the experiment, and $[\hat{m}'_i]_1$ are uniquely determined for $i \in \{0,1\}$. Therefore D is distributed identically to $\mathcal{A}$'s view in this case.

If on the other hand $\forall i \in \{0,1\} : \mathsf{Aggregate}((r_i, \mathbf{m}_0), S_i, (\hat{\sigma}_{i,j})_{j\in S_i}) \neq \perp$, then S sets $s_1, s_2 \leftarrow_{\mathrm{R}} \mathbb{Z}_p$ and outputs $\sigma_i = ([x + s_i(w + \langle \mathbf{m}_i, \mathbf{u}\rangle)]_1, [s_i]_1$ for $i \in \{0,1\}$. Since $\mathsf{VerifyPK}(pk) = 1$ we have that $w = w'$, $v = v'$, $\mathbf{u} = \mathbf{u}'$ and $\mathbf{y} = v\mathbf{u}$. Therefore the σ_i are valid signatures on $\mathbf{m}_i$ that are distributed identically to outputs of $\mathsf{Aggregate}$. As mentioned above, $[\hat{m}_i]_1$ and $[\hat{m}'_i]_1$ for $i \in \{0,1\}$ are distributed identically to $\mathcal{A}$'s view. Therefore D is distributed identically to $\mathcal{A}$'s view in this case.

Since S is independent of b we get that the scheme is perfectly blind. $\square$

5.2 Unforgeability

We follow the same approach laid out in Section 4.2, where now there are additional random exponents $\mathbf{a}$ (the randomness used by Share) which will be represented by the indeterminates $\mathbf{A}$. For the remainder of this section, let $C \subseteq [n]$ denote the set of (adaptively) corrupted issuers and let $\rho : [q] \to [n]$ map the i-th signing query to its issuer $\rho(i)$.

Step 1: Symbolic Security. The polynomials x_i correspond to the shares $[x_i]_1$ of the secret key $[x]_1$ with randomness $\mathbf{A}$. Since the adversary will know the shares $[x_c]_1$ for $c \in C$ we introduce some shorthand notation: For $i \le n$ define the polynomial $x_i \in \mathbb{Z}_p[X, \mathbf{A}]$ as

$$x_i := X + \sum_{k=1}^{t-1} \mathbf{A}[k] \cdot i^k. \tag{5}$$

For a set $C \subseteq [n]$ let $\mathbf{x}_C := (x_c)_{c\in C}$.

For $i \le q$ let $\hat{m}_i \in \mathbb{Z}_p[X, \mathbf{A}, W, \mathbf{U}, V, S_1, \ldots, S_q]$ and define the polynomial $z_i \in \mathbb{Z}_p[X, \mathbf{A}, W, \mathbf{U}, V, S_1, \ldots, S_q]$ as

$$z_i := x_{\rho(i)} + S_i(W + \hat{m}_i). \tag{6}$$

Lemma 5 (Structure of $\hat{m}_i$). *Let $C \subseteq [n]$ and let $\hat{m}_i, \hat{m}'_i \in \mathsf{span}\{1, \mathbf{x}_C, W, \mathbf{U}, V, V\mathbf{U}, S_1, \ldots, S_{i-1}, z_1, \ldots, z_{i-1}\}$. If $v \cdot \hat{m}_i \equiv \hat{m}'_i$, then $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$.*

Proof. We proceed by induction on $i \in [q]$.

<u>*Base case.*</u> Here, $V \cdot \hat{m}_1 \equiv \hat{m}'_1 \in \mathsf{span}\{1, \mathbf{x}_C, W, \mathbf{U}, V, V\mathbf{U}\}$. The only multiples of V appearing on the RHS are $V, V\mathbf{U}$. Therefore $V \cdot \hat{m}_1 \in \mathsf{span}\{V, V\mathbf{U}\}$ and thus $\hat{m}_1 \in \mathsf{span}\{1, \mathbf{U}\}$.

<u>*Induction step.*</u> By the induction hypothesis $\hat{m}_1, \ldots, \hat{m}_{i-1} \in \mathsf{span}\{1, \mathbf{U}\}$, implying

$$z_1, \ldots, z_{i-1} \in \mathsf{span}\{x_1, \ldots, x_n, S_1, \ldots, S_{i-1}, S_1 W, \ldots, S_{i-1} W, S_i \mathbf{U}, \ldots, S_{i-1}\mathbf{U}\}.$$

In particular, none of the monomials of $z_1, \ldots, z_{i-1}$ are multiples of V. Also we have

$$V \cdot \hat{m}_i \equiv \hat{m}'_i \in \mathsf{span}\{1, \mathbf{x}_C, W, \mathbf{U}, V, V\mathbf{U}, S_1, \ldots, S_{i-1}, z_1, \ldots, z_{i-1}\}.$$

Again, the only multiples of V appearing on the RHS are V and $V\mathbf{U}$. Therefore $V \cdot \hat{m}_i \in \mathsf{span}\{V, V\mathbf{U}\}$ and thus $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$. □

Lemma 6 (Structure of forgeries). *Let $C \subseteq [n]$ and let x_i be defined as in (5). Let z_i be defined as in (6) where $\hat{m}_i = r_i + \langle \mathbf{m}_i, \mathbf{U}\rangle$ for $r_i \in \mathbb{Z}_p$ and $\mathbf{m}_i \in \mathbb{Z}_p^\ell$. Suppose the polynomials $z^*, s^* \in \mathsf{span}\{1, \mathbf{x}_C, W, \mathbf{U}, V, V\mathbf{U}, S_1, \ldots, S_q, z_1, \ldots, z_q\}$ and the vector $\mathbf{m}^* \in \mathbb{Z}_p^\ell$ satisfy the polynomial equality*

$$X + s^*\left(W + \langle \mathbf{m}^*, \mathbf{U}\rangle\right) \equiv z^*. \tag{7}$$

Then there exists $J \subseteq [q]$ such that $|\rho(J) \cup C| \ge t$, and we have $\mathbf{m}_j = \mathbf{m}^*$ for all $j \in J$.

Proof. Let $\boldsymbol{\theta}, \boldsymbol{\eta} \in \mathbb{Z}_p^{2\ell+3}$ and $\boldsymbol{\psi}, \boldsymbol{\varphi} \in \mathbb{Z}_p^{|C|}$ and $\alpha_i, \beta_i, \gamma_i, \delta_i \in \mathbb{Z}_p$ for $i \in [q]$ be such that

$$\begin{aligned} z^* &:= \langle \boldsymbol{\theta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U}\rangle + \langle \boldsymbol{\varphi}, \mathbf{x}_C\rangle + \sum_i \alpha_i S_i + \beta_i z_i, \\ s^* &:= \langle \boldsymbol{\eta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U}\rangle + \langle \boldsymbol{\psi}, \mathbf{x}_C\rangle + \sum_i \gamma_i S_i + \delta_i z_i, \end{aligned} \tag{8}$$

where now $z_i = x_{\rho(i)} + S_i W + S_i r_i + \langle \mathbf{m}_i, S_i \mathbf{U}\rangle$. Expanding (7) by (8) yields

$$\begin{aligned} X + \Big(\langle \boldsymbol{\eta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U}\rangle + \langle \boldsymbol{\psi}, \mathbf{x}_C\rangle + \sum_i \gamma_i S_i + \delta_i z_i\Big)\big(W + \langle \mathbf{m}^*, \mathbf{U}\rangle\big) \\ = \langle \boldsymbol{\theta}, 1 \| W \| V \| \mathbf{U} \| V\mathbf{U}\rangle + \langle \boldsymbol{\varphi}, \mathbf{x}_C\rangle + \sum_i \alpha_i S_i + \beta_i z_i. \end{aligned} \tag{9}$$

Comparing coefficients of $S_i W^2$ yields:

$$\delta_1 = \cdots = \delta_q = 0.$$

Considering only scalar multiples of X, $\mathbf{A}$, $S_i W$, $S_i \mathbf{U}$ (i.e., comparing coefficients of these monomials) in (9) yields:

$$X + \big(\sum_i \gamma_i S_i\big)\big(W + \langle \mathbf{m}^*, \mathbf{U}\rangle\big) = \langle \boldsymbol{\varphi}, \mathbf{x}_C\rangle + \sum_i \beta_i \big(x_{\rho(i)} + S_i W + \langle \mathbf{m}_i, S_i \mathbf{U}\rangle\big).$$

Comparing coefficients of $S_i W$ yields $\gamma_i = \beta_i$. With this, comparing coefficients of $S_i \mathbf{U}$ yields $\beta_i \mathbf{m}^* = \beta_i \mathbf{m}_i$. Therefore

$$\beta_i = 0 \vee \mathbf{m}^* = \mathbf{m}_i \quad \forall i \in [q],$$

and comparing coefficients of X and $\mathbf{A}$ leaves

$$X = \langle \boldsymbol{\varphi}, \mathbf{x}_C\rangle + \sum_i \beta_i x_{\rho(i)}.$$

Let $J = \{i \in [q] : \mathbf{m}_i = \mathbf{m}^*\}$ then since $\beta_i = 0$ for $i \notin J$, we get

$$X = \langle \boldsymbol{\varphi}, \mathbf{x}_C\rangle + \sum_{i \in J} \beta_i x_{\rho(i)}.$$

Enumerate $C = \{c_1, \ldots, c_{|C|}\}$, then by (5) we have

$$X = \sum_{i \le |C|} \boldsymbol{\varphi}[i]\Big(X + \sum_{k=1}^{t-1} \mathbf{A}[k] c_i^k\Big) + \sum_{i \in J} \beta_i \Big(X + \sum_{k=1}^{t-1} \mathbf{A}[k] \rho(i)^k\Big).$$

Equating coefficients for X and $\mathbf{A}$ we get a non-homogenous system of t linear equations in variables $\boldsymbol{\varphi}$ and β_i for $i \in C$:

$$\begin{aligned} \sum_{i \le |C|} \boldsymbol{\varphi}[i] + \sum_{i \in J} \beta_i &= 1, \\ \sum_{i \le |C|} \boldsymbol{\varphi}[i] c_i^k + \sum_{i \in J} \beta_i \rho(i)^k &= 0 \qquad \text{for } k \in [t-1]. \end{aligned} \tag{10}$$

We regroup the variables that have the same coefficient. For $j \in [n]$ define

$$\chi_j := \sum_{i : c_i = j} \boldsymbol{\varphi}[i] + \sum_{i \in J : \rho(i) = j} \beta_i,$$

and set $\boldsymbol{\chi} := (\chi_1, \ldots, \chi_n)$. Note that the number of non-zero components of $\boldsymbol{\chi}$ is at most $|\rho(J) \cup C|$.

Towards a contradiction, assume that $|\rho(J) \cup C| < t$. Then there are indices $j_1, \ldots, j_{t-1}$ so that $\boldsymbol{\chi}$ is zero anywhere else. This means that the system of equations in (10) can be written as

$$\begin{pmatrix} \chi_{j_1} & \chi_{j_2} & \cdots & \chi_{j_{t-1}} \end{pmatrix} \cdot \begin{pmatrix} 1 & j_1 & j_1^2 & \cdots & j_1^{t-1} \\ 1 & j_2 & j_2^2 & \cdots & j_2^{t-1} \\ \vdots & \vdots & & \ddots & \\ 1 & j_{t-1} & j_{t-1}^2 & \cdots & j_{t-1}^{t-1} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Since t rows of a Vandermonde matrix of width t are linearly independent and the RHS together with the matrix above constitute a Vandermonde matrix $\mathbf{V}(0, j_1, \ldots, j_{t-1})$, the system of equations does not have a solution. We thus have $|\rho(J) \cup C| \ge t$, which completes the proof. $\square$

Step 2: Moving to the AGM. Provided these technical lemmata, we are now able to tackle unforgeability of our threshold blind signature scheme in the AGM.

Theorem 6 (Unforgeability in the AGM). *Let grp be a bilinear group of order p. Let $\mathcal{A}$ be an algebraic adversary against* tOMUF *security of the threshold blind signature scheme Σ for grp defined in Fig. 6 making q signature queries in total and running in time τ. Then there exists an algorithm $\mathcal{R}$ solving $(2,1)$-DL for grp, running in time at most $\tau + \mathcal{O}(q \log p)$ such that*

$$\mathsf{Adv}^{\mathsf{tOMUF}}_{\Sigma,\mathcal{A}} \leq \mathsf{Adv}^{(2,1)\text{-DL}}_{\mathcal{R}} + \frac{3}{p}.$$

Proof. Assume that $\mathcal{A}$ makes q *valid* signature queries $[\hat{m}]_1, [\hat{m}']_1$, that is, queries for which $[\hat{m}']_1 = v \cdot [\hat{m}]_1$. This is without loss of generality as for any adversary $\mathcal{A}$ there exists an adversary $\mathcal{A}'$ that behaves like $\mathcal{A}$ but checks whether a query is valid by computing $e\big([\hat{m}']_1, [1]_2\big) \stackrel{?}{=} e\big([\hat{m}]_1, [v]_2\big)$ and discards the query if it is not valid.

The proof will proceed as follows: first we describe how the reduction $\mathcal{R}$ chooses parameters and answers Signer queries by embedding the $(2,1)$-DL solution y into the logarithms of group elements. We will then analyze two cases depending on $\mathcal{A}$'s queries, where in each case $\mathcal{R}$ will use $\mathcal{A}$ to derive a polynomial h that is non-zero with high probability, such that $h(y) = 0$. In this case, the reduction can solve for y by computing the roots of h. At last we fully describe $\mathcal{R}$.

Consider a $(2,1)$-DL challenge $[1]_1, [1]_2, [y]_1, [y^2]_1, [y]_2$ given to $\mathcal{R}$. To embed the solution y into the public parameters, the public key, the shares of the secret key, and the randomness chosen in Signer, the reduction hides y as follows:

- For the public parameters and public key, $\mathcal{R}$ samples $\tilde{x}, \bar{x}, \tilde{v}, \bar{v}, \tilde{w}, \bar{w} \leftarrow_{\mathrm{R}} \mathbb{Z}_p$, $\tilde{\mathbf{u}}, \bar{\mathbf{u}} \leftarrow_{\mathrm{R}} \mathbb{Z}_p^{\ell}$ and implicitly defines:

  $$x = \bar{x} + \tilde{x} \cdot y, \quad v = \bar{v} + \tilde{v} \cdot y, \quad w = \bar{w} + \tilde{w} \cdot y, \quad \mathbf{u} = \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y.$$

  Since $\bar{x}$ was chosen uniformly, the resulting public key is distributed identically as in the security game. Analogously, the other parameters are distributed identically. $\mathcal{R}$ additionally samples $\tilde{\mathbf{a}}, \bar{\mathbf{a}} \leftarrow_{\mathrm{R}} \mathbb{Z}_p^{t-1}$ and sets:

  $$\mathbf{a} = \bar{\mathbf{a}} + \tilde{\mathbf{a}} \cdot y.$$

  The reduction simulates Shamir secret shares using the random values $\mathbf{a}$. For $f(Z) := x + \sum_{i=1}^{t-1} \mathbf{a}[i] Z^i$, define $x_j := f(j)$ and set $[x_j]_1$ as the secret key of party j. Note that for any $I \subseteq [n]$ with $|I| \geq t$ we have that $x = \sum_{i \in I} \lambda_i x_i$ where λ_i denotes the i-th Lagrange coefficient for the set I. For any $I \subseteq [n]$ with $|I| = t-1$, when $\mathcal{A}$ knows $\mathbf{x}_I$ then since there exists exactly one share explaining each secret value x, then $\mathcal{A}$ has no information on x. Since

  $$\begin{aligned} x_j = f(j) &= x + \sum_{i=1}^{t-1} \mathbf{a}[i] j^i \\ &= \bar{x} + \tilde{x} \cdot y + \sum_{i=1}^{t-1} (\bar{\mathbf{a}}[i] + \tilde{\mathbf{a}}[i] \cdot y) j^i \end{aligned}$$

$$= \bar{x} + \sum_{i=1}^{t-1} \bar{\mathbf{a}}[i] j^i + \left(\tilde{x} + \sum_{i=1}^{t-1} \tilde{\mathbf{a}}[i] j^i\right) \cdot y,$$

the reduction can answer corruption queries, even if made adaptively, since $\mathcal{R}$ can use the $(2,1)$-DL challenge to output the corresponding group elements.

– Consider the i-th Signer query to issuer $\rho(i)$ that satisfies $\hat{m}_i = r_i + \langle \mathbf{m}_i, \mathbf{U} \rangle \in \mathsf{span}\{1, \mathbf{U}\}$ (recall that by this we mean that $\mathcal{A}$ gives an algebraic representation $(r_i, \mathbf{m}_i)$ of $[\hat{m}_i]_1$ in basis $[1]_1$, $[\mathbf{u}]_1$). The reduction samples $\tilde{s}_i, \bar{s}_i \leftarrow_{\mathrm{R}} \mathbb{Z}_p$ and programs

$$s_i = \bar{s}_i + \tilde{s}_i \cdot y.$$

It then computes

$$\begin{aligned} z_i &= x_{\rho(i)} + s_i (w + \hat{m}_i) \\ &= x_{\rho(i)} + s_i (w + r_i + \langle \mathbf{m}_i, \mathbf{u} \rangle) \\ &= \bar{x} + \sum_{k=1}^{t-1} \bar{\mathbf{a}}[k] \rho(i)^k + \left(\tilde{x} + \sum_{k=1}^{t-1} \tilde{\mathbf{a}}[k] \rho(i)^k\right) \cdot y + (\bar{s}_i + \tilde{s}_i \cdot y)\left(\bar{w} + \tilde{w} \cdot y + r_i + \langle \mathbf{m}_i, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y \rangle\right) \\ &= \bar{x} + \sum_{k=1}^{t-1} \bar{\mathbf{a}}[k] \rho(i)^k + \bar{s}_i \left(\bar{w} + r_i + \langle \mathbf{m}_i, \bar{\mathbf{u}} \rangle\right) \\ &\quad + \left(\tilde{x} + \sum_{k=1}^{t-1} \tilde{\mathbf{a}}[k] \rho(i)^k + \bar{s}_i \left(\tilde{w} + \langle \mathbf{m}_i, \tilde{\mathbf{u}} \rangle\right) + \tilde{s}_i \left(\bar{w} + r_i + \langle \mathbf{m}_i, \bar{\mathbf{u}} \rangle\right)\right) \cdot y + \tilde{s}_i \left(\tilde{w} + \langle \mathbf{m}_i, \tilde{\mathbf{u}} \rangle\right) \cdot y^2 \end{aligned}$$

and outputs $[z_i]_1, [s_i]_1$.

Again, since $\bar{s}$ was chosen uniformly, the resulting signatures are distributed identically as in the security game.

We thus showed how $\mathcal{R}$ can simulate the game to $\mathcal{A}$, answer corruption queries and answer signing queries whenever $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$. Next consider the case that $\mathcal{A}$ does not provide such a representation on all Signer queries.

Case 1. There exists some $i \in [q]$ for which $\hat{m}_i \notin \mathsf{span}\{1, \mathbf{U}\}$. Take the smallest i for which this is the case. By the above, the reduction is able to answer all previous $i-1$ signature queries. Next, using the representations $\mathcal{A}$ provides for $[\hat{m}_i]_1, [\hat{m}'_i]_1$, the reduction can compute a polynomial $P_1 \in \mathbb{Z}_p[X, \mathbf{A}, W, \mathbf{U}, V, S_1, \ldots, S_{i-1}]$ corresponding to $V \cdot \hat{m}_i - \hat{m}'_i$. In particular let $C \subseteq [n]$ be the set of corrupted issuers at the i-th Signer query. Then the elements $\mathcal{A}$ previously received comprise $[1]_1, [\mathbf{x}_C]_1, [w]_1, [\mathbf{u}]_1, [v]_1, [v\mathbf{u}]_1$ and the elements $[s_j]_1, [z_j]_1$ for $j < i$. For a representation

$$[\hat{m}_i]_1 = \alpha[1]_1 + \langle \boldsymbol{\beta}, [\mathbf{x}_C]_1 \rangle + \gamma[w]_1 + \langle \boldsymbol{\delta}, [\mathbf{u}]_1 \rangle + \zeta[v]_1 + \langle \boldsymbol{\eta}, [v\mathbf{u}]_1 \rangle + \sum_{j<i} \theta_j [s_j]_1 + \kappa_j [z_j]_1,$$

provided by $\mathcal{A}$, the reduction defines the polynomial $\hat{m}_i \in \mathbb{Z}_p[X, \mathbf{A}, W, \mathbf{U}, V, S_1, \ldots, S_{i-1}]$ as

$$\hat{m}_i := \alpha + \langle \boldsymbol{\beta}, \mathbf{x}_C \rangle + \gamma W + \langle \boldsymbol{\delta}, \mathbf{U} \rangle + \zeta V + \langle \boldsymbol{\eta}, V\mathbf{U} \rangle + \sum_{j<i} \theta_j S_j + \kappa_j \left(X + S_j (W + \hat{m}_j)\right).$$

The polynomial $\hat{m}'_i$ is defined analogously. Since for all $j < i$, $\hat{m}_j \in \mathsf{span}\{1, \mathbf{U}\}$ we have that $\deg \hat{m}_j \leq 1$. Therefore $\deg \hat{m}_i \leq 2$ and $\deg \hat{m}'_i \leq 2$. The polynomial $P_1 := V \cdot \hat{m}_i - \hat{m}'_i$ has thus the following properties:

(i) P_1 has total degree (at most) 3.
(ii) P_1 is a non-zero polynomial, since, by Lemma 5, $V \cdot \hat{m}_i \equiv \hat{m}'_i$ would imply $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$.
(iii) $P_1(\bar{x} + \tilde{x} \cdot y, \bar{\mathbf{a}} + \tilde{\mathbf{a}} \cdot y, \bar{w} + \tilde{w} \cdot y, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y, \bar{v} + \tilde{v} \cdot y, \bar{s}_1 + \tilde{s}_1 \cdot y, \ldots, \bar{s}_{i-1} + \tilde{s}_{i-1} \cdot y) = 0$, since $v \cdot [\hat{m}_i]_1 = [\hat{m}'_i]_1$ for a valid query.

Next, the reduction defines the polynomial $h \in \mathbb{Z}_p[Z]$ given by

$$h(Z) = P_1(\bar{x} + \tilde{x} \cdot Z, \bar{\mathbf{a}} + \tilde{\mathbf{a}} \cdot Z, \bar{w} + \tilde{w} \cdot Z, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot Z, \bar{v} + \tilde{v} \cdot Z, \bar{s}_1 + \tilde{s}_1 \cdot Z, \ldots, \bar{s}_{i-1} + \tilde{s}_{i-1} \cdot Z).$$

By Lemma 2 and (ii), the leading coefficient of h is a non-zero polynomial in $\mathbb{Z}_p[\tilde{X}, \tilde{\mathbf{A}}, \tilde{W}, \tilde{\mathbf{U}}, \tilde{V}, \tilde{S}_1, \ldots, \tilde{S}_{i-1}]$ of degree at most 3. Since $\tilde{x}, \tilde{\mathbf{a}}, \tilde{w}, \tilde{\mathbf{u}}, \tilde{v}, \tilde{s}_1, \ldots, \tilde{s}_{i-1}$ are independent and perfectly hidden from the view of $\mathcal{A}$, applying the Schwartz-Zippel lemma (Lemma 1) yields that h is non-zero with probability at least $1 - 3/p$. From (iii), we know that $h(y) = 0$. Moreover, due to (i), h has degree at most 3. Therefore $\mathcal{R}$ can solve for y by computing the roots of the polynomial h.

Next consider how $\mathcal{R}$ behaves when Case 1 does not occur. We already established that $\mathcal{R}$ can then answer Signer queries correctly. Let's therefore consider that $\mathcal{A}$ wins the tOMUF game.

Case 2. $\hat{m}_1, \ldots, \hat{m}_q \in \mathsf{span}\{1, \mathbf{U}\}$ and $\mathcal{A}$ wins tOMUF. Let $\hat{m}_i = r_i + \langle \mathbf{m}_i, \mathbf{u} \rangle$ and suppose $\mathcal{A}$ provides valid signatures on k distinct messages $\mathbf{m}_1^*, \ldots, \mathbf{m}_k^*$. Let $J_1, \ldots, J_k \subseteq [q]$ denote the subset of Signer queries that correspond to the messages $\mathbf{m}_1^*, \ldots, \mathbf{m}_k^*$, that is $\mathbf{m}_j = \mathbf{m}_i^*$ for $j \in J_i$. Note that since the messages are distinct, we have that the J_i are pairwise disjoint.

If $|\rho(J_i) \cup C| \geq t$ for all $i \in [k]$, then by the definition of count we have $k \leq \mathsf{count}(\rho, t, C)$ and thus $\mathcal{A}$ did not win tOMUF. Therefore there exists i such that $|\rho(J_i) \cup C| < t$. Let $([z^*]_1, [s^*]_1)$ denote the signature for $\mathbf{m}_i^*$. Using the representations $\mathcal{A}$ provides for $[z^*]_1, [s^*]_1$, the reduction can compute a polynomial $P_2 \in \mathbb{Z}_p[X, \mathbf{A}, W, \mathbf{U}, V, S_1, \ldots, S_q]$ corresponding to $X + s^*(W + \langle \mathbf{m}_i^*, \mathbf{U} \rangle) - z^*$ with the following properties:

(i) P_2 has total degree (at most) 3. This is due to the fact that (since we assume Case 2) both s^* and z^* have total degree (at most) 2.
(ii) P_2 is a non-zero polynomial, due to Lemma 6 and $|\rho(J_i) \cup C| < t$.
(iii) $P_2(\bar{x} + \tilde{x} \cdot y, \bar{\mathbf{a}} + \tilde{\mathbf{a}} \cdot y, \bar{w} + \tilde{w} \cdot y, \bar{\mathbf{u}} + \tilde{\mathbf{u}} \cdot y, \bar{v} + \tilde{v} \cdot y, \bar{s}_1 + \tilde{s}_1 \cdot y, \ldots, \bar{s}_q + \tilde{s}_q \cdot y) = 0$, since $([z^*]_1, [s^*]_1)$ is a valid signature.

We proceed as in Case 1 and define a polynomial h that due to (ii) is non-zero with probability at least $1 - 3/p$, due to (i) has degree at most 3 and due to (iii) satisfies $h(y) = 0$. The reduction can then solve for y by factoring h.

The reduction $\mathcal{R}$. We will now describe $\mathcal{R}$ fully. It runs $\mathcal{A}$ in the tOMUF game and proceeds as follows:

- It computes pp, pk, and $sk_1, \ldots, sk_n$ as described above, and sends pp and pk to $\mathcal{A}$.
- When $\mathcal{A}$ sends a corruption query on $j \notin C$, $\mathcal{R}$ answers with sk_j and updates $C := C \cup \{j\}$.
- For each signature query $([\hat{m}_i]_1, [\hat{m}'_i]_1)$ to issuer j, it checks if the representation $\mathcal{A}$ provides for $[\hat{m}_i]_1$ satisfies $\hat{m}_i \in \mathsf{span}\{1, \mathbf{U}\}$. If so, it answers the query with $([z]_i, [y]_i)$ as described above. If not, it aborts and proceeds as in Case 1 to solve for y.

– When $\mathcal{A}$ outputs valid signatures on k distinct messages with $k > \mathsf{count}(\rho, t, C)$, $\mathcal{R}$ proceeds as in Case 2 to solve for y.

We analyze the probability of $\mathcal{R}$ winning the $(2,1)$-DL game. Let W denote the event that $\mathcal{R}$ wins the $(2,1)$-DL experiment, and C_1, C_2 denote the events that Case 1 or Case 2 occurs respectively. Since $\Pr[C_1] + \Pr[C_2] = 1$ we showed

$$\begin{aligned}
\mathsf{Adv}_{\mathcal{R}}^{(2,1)\text{-DL}} &= \Pr[W \mid C_1] \Pr[C_1] + \Pr[W \mid C_2] \Pr[C_2] \\
&\geq (1 - \tfrac{3}{p}) \Pr[C_1] + (1 - \tfrac{3}{p}) \mathsf{Adv}_{\Sigma,\mathcal{A}}^{\mathsf{tOMUF}} \Pr[C_2] \\
&\geq (1 - \tfrac{3}{p}) \mathsf{Adv}_{\Sigma,\mathcal{A}}^{\mathsf{tOMUF}} \\
&\geq \mathsf{Adv}_{\Sigma,\mathcal{A}}^{\mathsf{tOMUF}} - \tfrac{3}{p}.
\end{aligned}$$

$\mathcal{R}$ runs $\mathcal{A}$, which takes time τ, it answers q signing queries, up to $t-1$ corruption queries and picks public parameters and the public key in time $\mathcal{O}(q \log p)$, and it factors a polynomial of degree at most 3 in time $\mathcal{O}(\log p)$. In total, $\mathcal{R}$'s running time is in $\tau + \mathcal{O}(q \log p)$. □

The reader might wonder why it was necessary for $\mathcal{R}$ to run $\mathcal{A}$, even though it knew the secret key $[x]_1$. This is because in order to find the $(2,1)$-DL solution y the reduction $\mathcal{R}$ needs to obtain a forgery whose representation is independent of $[x]_1$. A winning adversary $\mathcal{A}$ guarantees such a forgery, whereas honestly generated signatures do not.

Acknowledgments. AI tools were used as typing assistants for grammar and basic editing. We thank the anonymous reviewers for valuable comments and pointing us to [HK17]. This work was funded by the Vienna Science and Technology Fund (WWTF) [10.47379/VRG18002] and [10.47379/ICT25-075] and by the Austrian Science Fund (FWF) [10.55776/F8515-N]. Part of this work was done while H.W. was visiting the Simons Institute for the Theory of Computing, supported in part by a grant from the UC Noyce Initiative.

References

ABGW17. Miguel Ambrona, Gilles Barthe, Romain Gay, and Hoeteck Wee. Attribute-based encryption in the generic group model: Automated proofs and new constructions. In Bhavani M. Thuraisingham, David Evans, Tal Malkin, and Dongyan Xu, editors, *ACM CCS 2017*, pages 647–664. ACM Press, October / November 2017.

AJOR18. Masayuki Abe, Charanjit S. Jutla, Miyako Ohkubo, and Arnab Roy. Improved (almost) tightly-secure simulation-sound QA-NIZK with applications. In Thomas Peyrin and Steven Galbraith, editors, *ASIACRYPT 2018, Part I*, volume 11272 of *LNCS*, pages 627–656. Springer, Cham, December 2018.

AMMM18. Shashank Agrawal, Peihan Miao, Payman Mohassel, and Pratyay Mukherjee. PASTA: PASsword-based threshold authentication. In David Lie, Mohammad Mannan, Michael Backes, and XiaoFeng Wang, editors, *ACM CCS 2018*, pages 2042–2059. ACM Press, October 2018.

App. Apple. iCloud private relay. Available at `https://www.apple.com/privacy/docs/iCloud_Private_Relay_Overview_Dec2021.PDF`.

BB04. Dan Boneh and Xavier Boyen. Efficient selective-ID secure identity based encryption without random oracles. In Christian Cachin and Jan Camenisch, editors, *EUROCRYPT 2004*, volume 3027 of *LNCS*, pages 223–238. Springer, Berlin, Heidelberg, May 2004.

BBS04. Dan Boneh, Xavier Boyen, and Hovav Shacham. Short group signatures. In Matthew Franklin, editor, *CRYPTO 2004*, volume 3152 of *LNCS*, pages 41–55. Springer, Berlin, Heidelberg, August 2004.

BCC04. Ernest F. Brickell, Jan Camenisch, and Liqun Chen. Direct anonymous attestation. In Vijayalakshmi Atluri, Birgit Pfitzmann, and Patrick McDaniel, editors, *ACM CCS 2004*, pages 132–145. ACM Press, October 2004.

BCKL09. Mira Belenkiy, Melissa Chase, Markulf Kohlweiss, and Anna Lysyanskaya. Compact e-cash and simulatable VRFs revisited. In Hovav Shacham and Brent Waters, editors, *PAIRING 2009*, volume 5671 of *LNCS*, pages 114–131. Springer, Berlin, Heidelberg, August 2009.

BFL20. Balthazar Bauer, Georg Fuchsbauer, and Julian Loss. A classification of computational assumptions in the algebraic group model. In Daniele Micciancio and Thomas Ristenpart, editors, *CRYPTO 2020, Part II*, volume 12171 of *LNCS*, pages 121–151. Springer, Cham, August 2020.

BFP21. Balthazar Bauer, Georg Fuchsbauer, and Antoine Plouviez. The one-more discrete logarithm assumption in the generic group model. In Mehdi Tibouchi and Huaxiong Wang, editors, *ASIACRYPT 2021, Part IV*, volume 13093 of *LNCS*, pages 587–617. Springer, Cham, December 2021.

BFQ21. Balthazar Bauer, Georg Fuchsbauer, and Chen Qian. Transferable E-cash: A cleaner model and the first practical instantiation. In Juan Garay, editor, *PKC 2021, Part II*, volume 12711 of *LNCS*, pages 559–590. Springer, Cham, May 2021.

BFR24. Balthazar Bauer, Georg Fuchsbauer, and Fabian Regen. On security proofs of existing equivalence class signature schemes. In Kai-Min Chung and Yu Sasaki, editors, *ASIACRYPT 2024, Part II*, volume 15485 of *LNCS*, pages 3–37. Springer, Singapore, December 2024.

BL13. Foteini Baldimtsi and Anna Lysyanskaya. Anonymous credentials light. In Ahmad-Reza Sadeghi, Virgil D. Gligor, and Moti Yung, editors, *ACM CCS 2013*, pages 1087–1098. ACM Press, November 2013.

BL22. Renas Bacho and Julian Loss. On the adaptive security of the threshold BLS signature scheme. In Heng Yin, Angelos Stavrou, Cas Cremers, and Elaine Shi, editors, *ACM CCS 2022*, pages 193–207. ACM Press, November 2022.

BLS01. Dan Boneh, Ben Lynn, and Hovav Shacham. Short signatures from the Weil pairing. In Colin Boyd, editor, *ASIACRYPT 2001*, volume 2248 of *LNCS*, pages 514–532. Springer, Berlin, Heidelberg, December 2001.

Bol03. Alexandra Boldyreva. Threshold signatures, multisignatures and blind signatures based on the gap-Diffie-Hellman-group signature scheme. In Yvo Desmedt, editor, *PKC 2003*, volume 2567 of *LNCS*, pages 31–46. Springer, Berlin, Heidelberg, January 2003.

BR93. Mihir Bellare and Phillip Rogaway. Random oracles are practical: A paradigm for designing efficient protocols. In Dorothy E. Denning, Raymond Pyle, Ravi Ganesan, Ravi S. Sandhu, and Victoria Ashby, editors, *ACM CCS 93*, pages 62–73. ACM Press, November 1993.

Bra94. Stefan Brands. Untraceable off-line cash in wallets with observers (extended abstract). In Douglas R. Stinson, editor, *CRYPTO'93*, volume 773 of *LNCS*, pages 302–318. Springer, Berlin, Heidelberg, August 1994.

CFN90. David Chaum, Amos Fiat, and Moni Naor. Untraceable electronic cash. In Shafi Goldwasser, editor, *CRYPTO'88*, volume 403 of *LNCS*, pages 319–327. Springer, New York, August 1990.

CG08. Jan Camenisch and Thomas Groß. Efficient attributes for anonymous credentials. In Peng Ning, Paul F. Syverson, and Somesh Jha, editors, *ACM CCS 2008*, pages 345–356. ACM Press, October 2008.

Cha82. David Chaum. Blind signatures for untraceable payments. In David Chaum, Ronald L. Rivest, and Alan T. Sherman, editors, *CRYPTO'82*, pages 199–203. Plenum Press, New York, USA, 1982.

Che06. Jung Hee Cheon. Security analysis of the strong Diffie-Hellman problem. In Serge Vaudenay, editor, *EUROCRYPT 2006*, volume 4004 of *LNCS*, pages 1–11. Springer, Berlin, Heidelberg, May / June 2006.

CKM+23. Elizabeth C. Crites, Chelsea Komlo, Mary Maller, Stefano Tessaro, and Chenzhi Zhu. Snowblind: A threshold blind signature in pairing-free groups. In Helena Handschuh and Anna Lysyanskaya, editors, *CRYPTO 2023, Part I*, volume 14081 of *LNCS*, pages 710–742. Springer, Cham, August 2023.

CL01. Jan Camenisch and Anna Lysyanskaya. An efficient system for non-transferable anonymous credentials with optional anonymity revocation. In Birgit Pfitzmann, editor, *EUROCRYPT 2001*, volume 2045 of *LNCS*, pages 93–118. Springer, Berlin, Heidelberg, May 2001.

CL03. Jan Camenisch and Anna Lysyanskaya. A signature scheme with efficient protocols. In Stelvio Cimato, Clemente Galdi, and Giuseppe Persiano, editors, *SCN 02*, volume 2576 of *LNCS*, pages 268–289. Springer, Berlin, Heidelberg, September 2003.

CL04. Jan Camenisch and Anna Lysyanskaya. Signature schemes and anonymous credentials from bilinear maps. In Matthew Franklin, editor, *CRYPTO 2004*, volume 3152 of *LNCS*, pages 56–72. Springer, Berlin, Heidelberg, August 2004.

Des88. Yvo Desmedt. Society and group oriented cryptography: A new concept. In Carl Pomerance, editor, *CRYPTO'87*, volume 293 of *LNCS*, pages 120–127. Springer, Berlin, Heidelberg, August 1988.

DF90. Yvo Desmedt and Yair Frankel. Threshold cryptosystems. In Gilles Brassard, editor, *CRYPTO'89*, volume 435 of *LNCS*, pages 307–315. Springer, New York, August 1990.

dPK22. Rafaël del Pino and Shuichi Katsumata. A new framework for more efficient round-optimal lattice-based (partially) blind signature via trapdoor sampling. In Yevgeniy Dodis and Thomas Shrimpton, editors, *CRYPTO 2022, Part II*, volume 13508 of *LNCS*, pages 306–336. Springer, Cham, August 2022.

DR24. Sourav Das and Ling Ren. Adaptively secure BLS threshold signatures from DDH and co-CDH. In Leonid Reyzin and Douglas Stebila, editors, *CRYPTO 2024, Part VII*, volume 14926 of *LNCS*, pages 251–284. Springer, Cham, August 2024.

FHKS16. Georg Fuchsbauer, Christian Hanser, Chethan Kamath, and Daniel Slamanig. Practical round-optimal blind signatures in the standard model from weaker assumptions. In Vassilis Zikas and Roberto De Prisco, editors, *SCN 16*, volume 9841 of *LNCS*, pages 391–408. Springer, Cham, August / September 2016.

FHS15. Georg Fuchsbauer, Christian Hanser, and Daniel Slamanig. Practical round-optimal blind signatures in the standard model. In Rosario Gennaro and Matthew J. B. Robshaw, editors, *CRYPTO 2015, Part II*, volume 9216 of *LNCS*, pages 233–253. Springer, Berlin, Heidelberg, August 2015.

FHS19. Georg Fuchsbauer, Christian Hanser, and Daniel Slamanig. Structure-preserving signatures on equivalence classes and constant-size anonymous credentials. *Journal of Cryptology*, 32(2):498–546, April 2019.

Fis05. Marc Fischlin. Communication-efficient non-interactive proofs of knowledge with online extractors. In Victor Shoup, editor, *CRYPTO 2005*, volume 3621 of *LNCS*, pages 152–168. Springer, Berlin, Heidelberg, August 2005.

Fis06. Marc Fischlin. Round-optimal composable blind signatures in the common reference string model. In Cynthia Dwork, editor, *CRYPTO 2006*, volume 4117 of *LNCS*, pages 60–77. Springer, Berlin, Heidelberg, August 2006.

FKL18. Georg Fuchsbauer, Eike Kiltz, and Julian Loss. The algebraic group model and its applications. In Hovav Shacham and Alexandra Boldyreva, editors, *CRYPTO 2018, Part II*, volume 10992 of *LNCS*, pages 33–62. Springer, Cham, August 2018.

FOO93. Atsushi Fujioka, Tatsuaki Okamoto, and Kazuo Ohta. A practical secret voting scheme for large scale elections. In Jennifer Seberry and Yuliang Zheng, editors, *AUSCRYPT'92*, volume 718 of *LNCS*, pages 244–251. Springer, Berlin, Heidelberg, December 1993.

FS87. Amos Fiat and Adi Shamir. How to prove yourself: Practical solutions to identification and signature problems. In Andrew M. Odlyzko, editor, *CRYPTO'86*, volume 263 of *LNCS*, pages 186–194. Springer, Berlin, Heidelberg, August 1987.

FS10. Marc Fischlin and Dominique Schröder. On the impossibility of three-move blind signature schemes. In Henri Gilbert, editor, *EUROCRYPT 2010*, volume 6110 of *LNCS*, pages 197–215. Springer, Berlin, Heidelberg, May / June 2010.

GG14. Sanjam Garg and Divya Gupta. Efficient round optimal blind signatures. In Phong Q. Nguyen and Elisabeth Oswald, editors, *EUROCRYPT 2014*, volume 8441 of *LNCS*, pages 477–495. Springer, Berlin, Heidelberg, May 2014.

Goo. Google. VPN by Google One. Available at `https://one.google.com/about/vpn/howitworks`.

GRS+11. Sanjam Garg, Vanishree Rao, Amit Sahai, Dominique Schröder, and Dominique Unruh. Round optimal blind signatures. In Phillip Rogaway, editor, *CRYPTO 2011*, volume 6841 of *LNCS*, pages 630–648. Springer, Berlin, Heidelberg, August 2011.

HIP+. Scott Hendrickson, Jana Iyengar, Tommy Pauly, Steven Valdez, and Christopher A. Wood. Private Access Tokens. Internet-Draft draft-private-access-tokens-00, Internet Engineering Task Force. Work in Progress.

HK17. Lucjan Hanzlik and Kamil Kluczniak. Two-move and setup-free blind signatures with perfect blindness. In *Proceedings of the 4th ACM International Workshop on ASIA Public-Key Cryptography*, APKC '17, page 1–11, New York, NY, USA, 2017. Association for Computing Machinery.

HKL19. Eduard Hauck, Eike Kiltz, and Julian Loss. A modular treatment of blind signatures from identification schemes. In Yuval Ishai and Vincent Rijmen, editors, *EUROCRYPT 2019, Part III*, volume 11478 of *LNCS*, pages 345–375. Springer, Cham, May 2019.

HLW23. Lucjan Hanzlik, Julian Loss, and Benedikt Wagner. Rai-choo! Evolving blind signatures to the next level. In Carmit Hazay and Martijn Stam, editors, *EUROCRYPT 2023, Part V*, volume 14008 of *LNCS*, pages 753–783. Springer, Cham, April 2023.

JKKX17. Stanislaw Jarecki, Aggelos Kiayias, Hugo Krawczyk, and Jiayu Xu. TOPPSS: Cost-minimal password-protected secret sharing based on threshold OPRF. In Dieter Gollmann, Atsuko Miyaji, and Hiroaki Kikuchi, editors, *ACNS 2017*, volume 10355 of *LNCS*, pages 39–58. Springer, Cham, July 2017.

JN25. Stanislaw Jarecki and Phillip Nazarian. Adaptively secure threshold blind BLS signatures and threshold oblivious PRF. In *ASIACRYPT 2025, Part VI*, LNCS, pages 601–634. Springer, Singapore, December 2025.

JX25. Jake Januzelli and Jiayu Xu. A complete characterization of one-more assumptions in the algebraic group model. In *ASIACRYPT 2025, Part IV*, LNCS, pages 206–238. Springer, Singapore, December 2025.

KMMM23. Aniket Kate, Easwar Vivek Mangipudi, Siva Maradana, and Pratyay Mukherjee. FlexiRand: Output private (distributed) VRFs and application to blockchains. In Weizhi Meng, Christian Damsgaard Jensen, Cas Cremers, and Engin Kirda, editors, *ACM CCS 2023*, pages 1776–1790. ACM Press, November 2023.

KNYY21. Shuichi Katsumata, Ryo Nishimaki, Shota Yamada, and Takashi Yamakawa. Round-optimal blind signatures in the plain model from classical and quantum standard assumptions. In Anne Canteaut and François-Xavier Standaert, editors, *EUROCRYPT 2021, Part I*, volume 12696 of *LNCS*, pages 404–434. Springer, Cham, October 2021.

KRS23. Shuichi Katsumata, Michael Reichle, and Yusuke Sakai. Practical round-optimal blind signatures in the ROM from standard assumptions. In Jian Guo and Ron Steinfeld, editors, *ASIACRYPT 2023, Part II*, volume 14439 of *LNCS*, pages 383–417. Springer, Singapore, December 2023.

Lip12. Helger Lipmaa. Progression-free sets and sublinear pairing-based non-interactive zero-knowledge arguments. In Ronald Cramer, editor, *Theory of Cryptography*, pages 169–189, Berlin, Heidelberg, 2012. Springer Berlin Heidelberg.

LNÖ25. Anja Lehmann, Phillip Nazarian, and Cavit Özbay. Stronger security for threshold blind signatures. In Serge Fehr and Pierre-Alain Fouque, editors, *EUROCRYPT 2025, Part II*, volume 15602 of *LNCS*, pages 335–364. Springer, Cham, May 2025.

OO92. Tatsuaki Okamoto and Kazuo Ohta. Universal electronic cash. In Joan Feigenbaum, editor, *CRYPTO'91*, volume 576 of *LNCS*, pages 324–337. Springer, Berlin, Heidelberg, August 1992.

PCM21. Click fraud prevention and attribution sent to advertiser, 2021. Available at `https://webkit.org/blog/11940/pcm-click-fraud-prevention-and-attribution-sent-to-advertiser/`.

PS16. David Pointcheval and Olivier Sanders. Short randomizable signatures. In Kazue Sako, editor, *CT-RSA 2016*, volume 9610 of *LNCS*, pages 111–126. Springer, Cham, February / March 2016.

SAB+19. Alberto Sonnino, Mustafa Al-Bassam, Shehar Bano, Sarah Meiklejohn, and George Danezis. Coconut: Threshold issuance selective disclosure credentials with applications to distributed ledgers. In *NDSS 2019*. The Internet Society, February 2019.

Tru. Trust tokens. Available at `https://developer.chrome.com/docs/privacy-sandbox/trust-tokens/`.

TZ23. Stefano Tessaro and Chenzhi Zhu. Revisiting BBS signatures. In Carmit Hazay and Martijn Stam, editors, *EUROCRYPT 2023, Part V*, volume 14008 of *LNCS*, pages 691–721. Springer, Cham, April 2023.

VZK03. Duc-Liem Vo, Fangguo Zhang, and Kwangjo Kim. A new threshold blind signature scheme from pairings. In *SCIS2003*, pages 233–238. SCIS, 2003.

A Equivalence of Definitions of count

Lehmann et al. [LNÖ25] use a different definition of count, which we will denote by $\mathsf{count}'_t : \mathbb{N}^n \mapsto \mathbb{N}$. We will show that the two definitions are equivalent. Let $V_{n,t} := \{\mathbf{v} \in \{0,1\}^n \mid \sum_i \mathbf{v}[i] = t\}$ be the set of length n binary sequences with hamming weight t. For $a, b \in \mathbb{N}^n$ let $a \le b :\Leftrightarrow \forall i : a[i] \le b[i]$ be defined component-wise. Let $\mathbf{q}$ be a vector such that the i-th component contains the number of signature queries made to the i-th issuer. Let $V_{n,t}(\mathbf{q}) := \{v \in V_{n,t} : v \le \mathbf{q}\}$ then count'_t is defined inductively:

$$\mathsf{count}'_t(\mathbf{q}) := \begin{cases} 1 + \max_{\mathbf{v} \in V_{n,t}(\mathbf{q})} \mathsf{count}'_t(\mathbf{q} - \mathbf{v}) & V_{n,t}(\mathbf{q}) \neq \emptyset \\ 0 & \text{otherwise.} \end{cases}$$

They remark that their definition is equivalent to one used in previous works [JKKX17, AMMM18, KMMM23]:

$$\mathsf{count}_t(\mathbf{q}) := \max\{k \mid \exists \mathbf{v}_1, \ldots, \mathbf{v}_k \in V_{n,t} : \mathbf{v}_1 + \cdots + \mathbf{v}_k \le \mathbf{q}\}.$$

We will now show that our definition is equivalent to this definition, and note that in their case, for a set of corrupted participants C, the winning condition of $\mathsf{tOMUF}_{\Sigma,\mathcal{A}}$ would consider $\mathsf{count}_{t-|C|}(\mathbf{q})$. Given $\rho : [q] \to [n]$ note that $\mathbf{q} = (|\rho^{-1}(1)|, \ldots, |\rho^{-1}(n)|)$, where ρ^{-1} denotes the preimage.

Let $k = \mathsf{count}(\rho, t, C)$ then there exist pairwise disjoint $J_1, \ldots, J_k$ such that $|\rho(J_i)| = t - |C|$, since whenever $|\rho(J'_i)| > t - |C|$ there exists J_i such that $|\rho(J_i)| = t - |C|$. Let $\mathbf{v}_i$ be the indicator vector of $\rho(J_i)$, then since J_i were pairwise disjoint, $\mathbf{v}_1 + \cdots + \mathbf{v}_k \le \mathbf{q}$ and therefore $\mathsf{count}(\rho, t, C) \le \mathsf{count}_{t-|C|}(\mathbf{q})$. On the other hand let $k = \mathsf{count}_{t-|C|}(\mathbf{q})$. Then there exist $\mathbf{v}_1, \ldots, \mathbf{v}_k$ from which we build $J_1, \ldots, J_k$ inductively: For each j such that $\mathbf{v}_i[j] = 1$ select an element $h \in \rho^{-1}(j)$ that previously was not selected. Since $\mathbf{v}_1 + \cdots + \mathbf{v}_k \le \mathbf{q}$ such an element always exists. By construction $J_1, \ldots, J_k$ are pairwise disjoint, and since $\mathbf{v}_i \in V_{n,t-|C|}$ we have $|\rho(J_i)| = t - |C|$, and therefore $\mathsf{count}(\rho, t, C) \ge \mathsf{count}_{t-|C|}(\mathbf{q})$.

B Proofs of Knowledge for Signatures

(Attribute-based) anonymous credentials enable a user, who possesses a signature on a vector of her attributes issued by some credential authority, to *selectively disclose* a subset of her attributes. This can be realized by giving a zero-knowledge proof of knowledge of the non-disclosed message components and a signature on the combined message. We follow the standard approach and define a Σ-protocol, for which we show *special soundness* and *(honest-verifier) zero knowledge*, which can then be compiled into a non-interactive protocol using the Fiat-Shamir transformation. Table 3 compares the efficiency of our protocol with prior works.

B.1 Definitions

For completeness we will define Σ-protocols below. In particular, we will consider Σ-protocols for the NP-relation

$$\mathsf{R} := \Big\{\big((pk, \hat{\mathbf{m}}, J), (\mathbf{m}, \sigma)\big) : \mathbf{m}[J] = \hat{\mathbf{m}} \wedge \mathsf{Verify}(pk, \mathbf{m}, \sigma) = 1\Big\}, \tag{11}$$

where $\mathbf{m} \in \mathbb{Z}_p^{\ell}$, $J \subseteq [\ell]$ and $\hat{\mathbf{m}} \in \mathbb{Z}_p^{|J|}$.

Definition 12. *A Σ-protocol for an NP-relation $\mathcal{R}$ with common input inst := $(pk, \hat{\mathbf{m}}, J)$ and private input wit := $(\mathbf{m}, \sigma)$ is a three-move interactive protocol between a prover and a verifier consisting of the algorithms $(\Sigma.\mathrm{P1}, \Sigma.\mathrm{P2}, \Sigma.\mathrm{C}, \Sigma.\mathrm{V})$ run as specified below:*

– *The prover takes the public input* inst *and its private input* wit*, runs* $(a, st) \leftarrow \Sigma.\mathrm{P1}(inst, wit)$ *and sends* a *to the verifier.*
– *The verifier runs* $c \leftarrow \Sigma.\mathrm{C}(inst)$ *and sends the challenge* c *to the prover.*
– *The prover sends* $s \leftarrow \Sigma.\mathrm{P2}(st, c)$ *to the verifier.*
– *The verifier outputs* $\Sigma.\mathrm{V}(inst, a, c, s) \in \{0, 1\}$.

Properties of a Σ-protocol are *completeness, special-soundness* and *(honest verifier) zero knowledge* as defined below.

Definition 13. *A* Σ*-protocol for* $\mathcal{R}$ *in* (11) *is* complete *if for all valid message-signature pair* $(\mathbf{m}, \sigma)$ *as private input* wit*, and all* $\hat{\mathbf{m}}$ *and* J *such that* $\hat{\mathbf{m}} = \mathbf{m}[J]$*, the verifier outputs* 1 *after an execution of the protocol.*

Definition 14. *A* Σ*-protocol is* special-sound *if there exists an efficient algorithm which given any two valid transcripts* (inst, a, c, s) *and* $(\mathrm{inst}, a, c', s')$ *with* $c \neq c'$ *can extract a valid witness* $(\mathbf{m}, \sigma)$ *such that* $\mathbf{m}[J] = \hat{\mathbf{m}}$*, where* J *and* $\hat{\mathbf{m}}$ *were defined in inst.*

Definition 15. *A* Σ*-protocol is* (honest-verifier) zero-knowledge *if there exists a simulator* S *such that the transcript of a honest execution is distributed identically to the output of* S.

B.2 Construction

Our Σ-protocol. The common inputs are the parameters $pp = ([\mathbf{u}]_1, [\mathbf{u}]_2, [w]_1, [w]_2)$ and a public key $pk = ([x]_T, [v]_1, [v]_2, [v\mathbf{u}]_1)$, a set $J \subseteq [\ell]$ of indices (whose complement we denote by $\overline{J} := [\ell] \setminus J$) and a family of disclosed message components $\hat{\mathbf{m}} := (\hat{\mathbf{m}}[i])_{i \in J}$. The protocol works as follows:

1. Given a signature $([z]_1, [s]_1)$ on $\mathbf{m}$ under pk as witness, such that $\mathbf{m}[J] = \hat{\mathbf{m}}$ (i.e. $\hat{\mathbf{m}}$ is a subfamily of $\mathbf{m}$), the prover samples $t, t', s' \leftarrow_{\mathrm{R}} \mathbb{Z}_p, \mathbf{m}' \leftarrow_{\mathrm{R}} \mathbb{Z}_p^{|\overline{J}|}$ and computes
$$\begin{aligned}[]
[\hat{s}]_1 &:= [s]_1 + [s']_1, \\
[\hat{z}]_1 &:= [z] + s'\big([w]_1 + \langle \mathbf{m}, [\mathbf{u}]_1 \rangle\big) + t[\hat{s}]_1, \\
R &:= e([\hat{s}]_1, \langle \mathbf{m}', [\mathbf{u}[\overline{J}]]_2 \rangle + [t']_2) \in \mathbb{G}_T,
\end{aligned}$$
and sends $[\hat{z}]_1, [\hat{s}]_1, R$ to the verifier.
2. The verifier sends a random challenge $c \leftarrow_{\mathrm{R}} \mathbb{Z}_p$ to the prover.
3. The prover replies with $\mathbf{m}'', t''$ where
$$\begin{aligned}
\mathbf{m}'' &:= \mathbf{m}' + c \cdot \mathbf{m}[\overline{J}], \\
t'' &:= t' + ct.
\end{aligned}$$
4. The verifier accepts if and only if
$$R \cdot e([\hat{z}], [c]_2) = [cx]_T \cdot e\big([\hat{s}]_1, \langle \mathbf{m}'', [\mathbf{u}[\overline{J}]]_2 \rangle + [t'']_2 + c([w]_2 + \langle \hat{\mathbf{m}}, [\mathbf{u}[J]]_2 \rangle)\big). \tag{12}$$

We note that the Σ-protocol would be more efficient if the prover sent $[r]_2 := \langle \mathbf{m}', [\mathbf{u}[\bar{J}]]_2\rangle + [t']_2$ instead of R (and in the verification equation, R was replaced by $e([\hat{s}]_1, [r]_2)$). Our variant however allows us to replace R by the challenge c (which for BLS12-381 is much shorter than $\mathbb{G}_2$ elements) in the Fiat-Shamir transform.

Lemma 7. *Our Σ-protocol is complete.*

Proof. Let $([z]_1, [s]_1)$ be a signature on $\mathbf{m}$. Let $J \subseteq [\ell]$ and $\hat{\mathbf{m}} \in \mathbb{Z}_p^{|J|}$ such that $\hat{\mathbf{m}} = \mathbf{m}[J]$. Since $[z]_1, [s]_1$ is valid on $\mathbf{m}$ we get $z = x + s(w + \langle \mathbf{m}, \mathbf{u}\rangle)$ and therefore $\hat{z} = x + \hat{s}(w + \langle \mathbf{m}, \mathbf{u}\rangle + t)$. Let r be the logarithm of R then $r = \hat{s}(\langle \mathbf{m}', \mathbf{u}[J]\rangle + t')$. Therefore the LHS of the verification equation (12) has logarithms

$$\begin{aligned}\hat{s}\big(\langle \mathbf{m}', \mathbf{u}[\bar{J}]\rangle + t'\big) + cx + c\hat{s}(w + \langle \mathbf{m}, \mathbf{u}\rangle + t)\\ = cx + \hat{s}\big(\langle \mathbf{m}', \mathbf{u}[\bar{J}]\rangle + t' + c(w + \langle \mathbf{m}, \mathbf{u}\rangle + t)\big),\end{aligned}$$

where by definition of $\mathbf{m}''$ and t'' and since $\langle \mathbf{m}, \mathbf{u}\rangle = \langle \mathbf{m}[\bar{J}], \mathbf{u}[\bar{J}]\rangle + \langle \mathbf{m}[J], \mathbf{u}[J]\rangle$ we have

$$= cx + \hat{s}\big(\langle \mathbf{m}'', \mathbf{u}[\bar{J}]\rangle + t'' + c(w + \langle \mathbf{m}[J], \mathbf{u}[J]\rangle)\big).$$

Since $\mathbf{m}[\bar{J}] = \hat{\mathbf{m}}$ this is the logarithm of the RHS, and the verifier accepts. □

Lemma 8. *Our Σ-protocol is special-sound.*

Proof. Let $J \subseteq [\ell]$. Let $c_1 \neq c_2$ and $(\mathbf{m}_1'', t_1'')$, $(\mathbf{m}_2'', t_2'')$ be two accepting transcripts whose first message is $([\hat{z}]_1, [\hat{s}]_1, R)$. We show how to extract a message-signature pair $(\mathbf{m}, \sigma)$ with $\mathbf{m}[J] = \hat{\mathbf{m}}$ such that σ is a valid signature for $\mathbf{m}$ under *pk*.

Let r be the logarithm of R. Since both transcripts are accepting, taking the logarithm of (12) we get

$$r + c_1\hat{z} = c_1 x + \hat{s}\big(\langle \mathbf{m}_1'', \mathbf{u}[\bar{J}]\rangle + t_1'' + c_1(w + \langle \hat{\mathbf{m}}, \mathbf{u}[J]\rangle)\big), \tag{13}$$

and

$$r + c_2\hat{z} = c_2 x + \hat{s}\big(\langle \mathbf{m}_2'', \mathbf{u}[\bar{J}]\rangle + t_2'' + c_2(w + \langle \hat{\mathbf{m}}, \mathbf{u}[J]\rangle)\big). \tag{14}$$

Then considering the difference (13) − (14) we get

$$(c_1 - c_2)\hat{z} = (c_1 - c_2)x + \hat{s}\big(\langle \mathbf{m}_1'' - \mathbf{m}_2'', \mathbf{u}[\bar{J}]\rangle + t_1'' - t_2'' + (c_1 - c_2)(w + \langle \hat{\mathbf{m}}, \mathbf{u}[J]\rangle)\big),$$

dividing both sides by $(c_1 - c_2) \neq 0$ we have

$$\hat{z} = x + \hat{s}\Big(\Big\langle \tfrac{\mathbf{m}_1'' - \mathbf{m}_2''}{c_1 - c_2}, \mathbf{u}[\bar{J}]\Big\rangle + \frac{t_1'' - t_2''}{c_1 - c_2} + w + \langle \hat{\mathbf{m}}, \mathbf{u}[J]\rangle\Big),$$

and therefore

$$\hat{z} - \hat{s}\tfrac{t_1'' - t_2''}{c_1 - c_2} = x + \hat{s}\Big(\Big\langle \tfrac{\mathbf{m}_1'' - \mathbf{m}_2''}{c_1 - c_2}, \mathbf{u}[\bar{J}]\Big\rangle + w + \langle \hat{\mathbf{m}}, \mathbf{u}[J]\rangle\Big). \tag{15}$$

Let $\mathbf{m}^*$ such that $\mathbf{m}^*[J] = \hat{\mathbf{m}}$ and $\mathbf{m}[\bar{J}] = \frac{\mathbf{m}_1'' - \mathbf{m}_2''}{c_1 - c_2}$, then by (15) we have that $[z^*]_1 := [\hat{z}]_1 - \frac{t_1'' - t_2''}{c_1 - c_2}[\hat{s}]_1$ and $[s^*]_1 := [\hat{s}]_1$ is a valid signature on $\mathbf{m}^*$ under *pk*.

From a fork of transcripts we can thus extract a valid signature on a message which has the disclosed components. This shows special (knowledge) soundness. □

Lemma 9. *Our* Σ*-protocol is (honest verifier) zero-knowledge.*

Proof. We will construct a simulator S such that the transcript of an honest execution is distributed identically to the output of S. The simulator S samples $[\hat{z}]_1, [\hat{s}]_1 \leftarrow \mathbb{G}_1$ and $c, t'' \leftarrow \mathbb{Z}_p$ and $\mathbf{m}'' \leftarrow \mathbb{Z}_p^k$ and sets

$$R := e\big([\hat{z}]_1, [-c]_2\big) \cdot [cx]_T \cdot e\Big([\hat{s}]_1,\ \langle \mathbf{m}'', [\mathbf{u}[\bar{J}]]_2\rangle + [t'']_2 + c\big([w]_2 + \langle \hat{\mathbf{m}}, [\mathbf{u}[J]]_2\rangle\big)\Big). \tag{16}$$

Since in the honest protocol $[\hat{z}]_1$ is uniform due to t, $[\hat{s}]_1$ is uniform due to s', $\mathbf{m}''$ is uniform due to $\mathbf{m}'$, t'' is uniform due to t' and R is the unique value in $\mathbb{G}_T$ that makes the verification equation true, the simulator's output is distributed identically to a honest execution. Therefore the protocol is zero-knowledge. □

Non-interactive zkPoK. Our protocol can be made non-interactive via the Fiat-Shamir (FS) transformation [FS87] by using a hash function $H : \{0,1\}^* \to \mathbb{Z}_p$. To prove the statement $\hat{\mathbf{m}}$, the prover computes $[\hat{z}]_1$,$[\hat{s}]_1$ and R as in the protocol, sets $c := H(pk, \hat{\mathbf{m}}, [\hat{z}]_1, [\hat{s}]_1, R)$ and computes $\mathbf{m}''$, t'' as in the protocol. The proof is then defined as

$$\big([\hat{z}]_1, [\hat{s}]_1, c, \mathbf{m}'', t''\big) \in \mathbb{G}_1^2 \times \mathbb{Z}_p^{k+2}$$

and is verified by first computing R as in (16) and accepting if and only if $c = H(pk, \hat{\mathbf{m}}, [\hat{z}]_1, [\hat{s}]_1, R)$.

Security of this construction follows from standard results about FS transforms [FS87] and the fact that the proof format $([\hat{z}]_1, [\hat{s}]_1, R, \mathbf{m}'', t'')$ can be efficiently transformed into ours and vice versa.